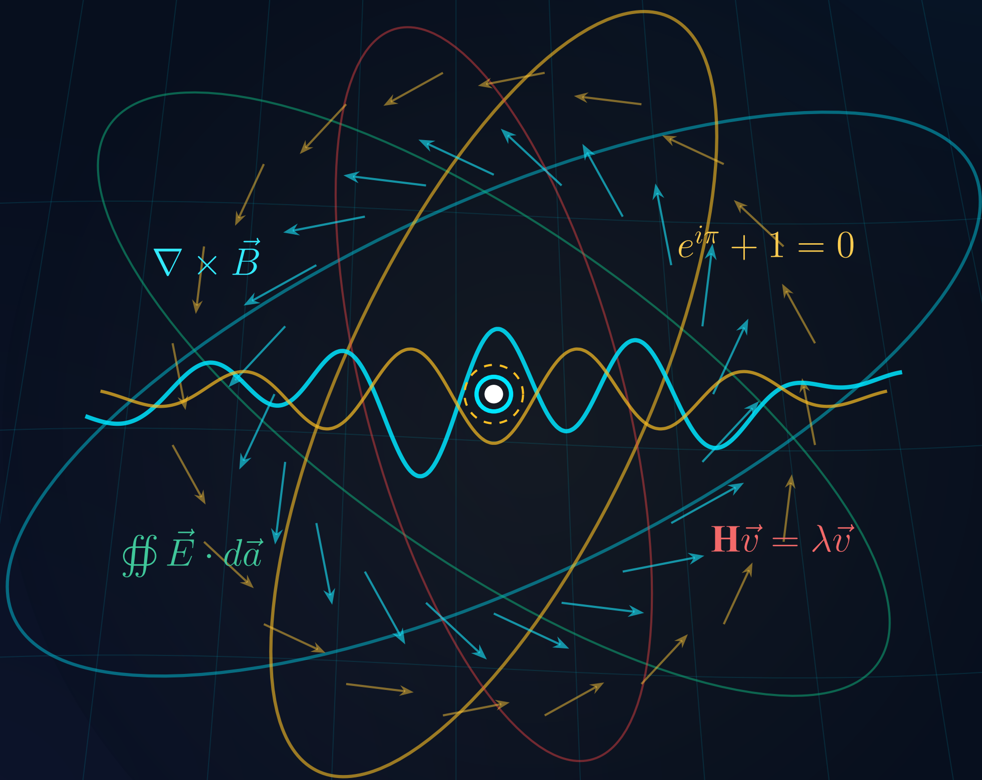
MATHEMATICS FOR PHYSICS

Analytical Foundations and Scientific Simulations

A Modern Academic Course for Scientists and Engineers

- Coordinate Systems
- Vector Calculus
- Infinite Series
- Complex Analysis
- Matrix Algebra
- Differential Equations
- Numerical Python Models

Featuring Crisp Vector Graphics and P-S-C-T Problem Solving Framework



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MATHEMATICS FOR PHYSICS

Analytical Foundations and Scientific Simulations

A Modern Pedagogical Textbook for Undergraduate and Graduate Physics

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Ph.D. (Applied Physics)

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Preface

Mathematics serves as the quintessential language of physics. From classical Newtonian trajectories to general relativity and quantum field theories, physical laws achieve precision, predictive power, and beauty through mathematical formulation.

This textbook, **Mathematics for Physics**, is designed specifically for undergraduate and beginning graduate students in physics, applied physics, and related engineering disciplines. The core mission of this book is to bridge the gap between abstract mathematical formalism and concrete physical intuition. Rather than treating mathematical theorems in isolation, every technique introduced herein is directly motivated by real-world physical systems—including electromagnetic field distributions, planetary motion, oscillating circuits, and quantum state evolutions.

Pedagogical Framework: The Active Learning Cycle

To maximize conceptual retention and analytical proficiency, each chapter integrates a structured four-stage pedagogical cycle:

1. **Conceptual Motivation:** Clear physical context establishing why a specific mathematical tool is necessary.

2. **Rigorous Analytical Development:** Thorough derivations that maintain mathematical integrity without sacrificing accessibility.

3. **P-S-C-T Worked Examples:** Problem-solving modeled through four explicit steps:

   - **Picture (P):** Physical conceptualization, coordinate selection, and free-body/field sketches.
   - **Strategy (S):** Identifying applicable physical principles, conservation laws, or symmetry arguments.
   - **Calculation (C):** Step-by-step mathematical execution with dimensional verification.
   - **Think (T):** Physical interpretation of limiting cases, asymptotic behaviors, and practical implications.

4. **Numerical Integration & Python Modeling:** Modern scientific computing scripts using NumPy, SciPy, and Matplotlib to visualize complex dynamics and solve differential equations beyond closed-form limits.

How to Use This Book

Chapters 1 through 2 establish the geometric and differential foundations of curvilinear coordinates and vector calculus. Chapters 3 and 4 explore infinite series expansions and complex analysis, vital for optics and wave mechanics. Chapter 5 develops matrix mechanics and eigenvalue techniques. Chapters 6 and 7 provide a rigorous treatment of multivariable calculus and the fundamental integral transformation theorems of Gauss, Stokes, and Green. Finally, Chapters 8 and 9 focus on first- and second-order ordinary differential equations, emphasizing damped, driven, and resonant physical oscillators.

I express my deepest appreciation to my colleagues in the Department of Physics at Rambhai Barni Rajabhat University, whose scholarly feedback has greatly enriched this work, and to the dedicated physics students whose insightful questions continue to shape its evolution.

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Course Syllabus & Instructional Design

Course Information

- **Course Code & Title:** PHYS2026 Mathematics for Physics
- **Credit Hours:** 3 Credits (3-0-6) [Lecture 3 hrs, Practical 0 hrs, Self-Study 6 hrs/week]
- **Curriculum:** Bachelor of Science in Physics, Core Major Requirement
- **Lead Instructor:** Asst. Prof. Dr. Chewa Thassana (chewa.t@rbru.ac.th)
- **Target Academic Level:** Year 2, Semester 1

Course Description

Advanced mathematical methods and computational techniques for physical applications. Topics include: curvilinear coordinate systems (Cartesian, cylindrical, spherical) and metric scale factors; vector analysis, gradient, divergence, curl, and conservative fields; Taylor and Maclaurin expansions, asymptotic approximations, and Fourier series; complex numbers, Euler's formula, phasors, and harmonic oscillations; matrix algebra, determinants, eigenvalues, eigenvectors, and diagonalization; multivariable differential calculus, tangent planes, directional derivatives, and Lagrange multipliers; multiple integrals, Jacobians, Gauss's Divergence theorem, and Stokes' theorem; analytical and numerical solutions of first-order and second-order ordinary differential equations with physical modeling in Python.

Course Learning Outcomes (CLOs) & Bloom's Taxonomy

Upon successful completion of this course, students will be able to:

1. **CLO 1 [Cognitive: Understanding / Remembering]:** Articulate foundational mathematical concepts and their direct correspondence to physical laws in classical mechanics, electromagnetism, and wave phenomena.

2. **CLO 2 [Cognitive: Applying / Analyzing]:** Formulate physical boundary-value problems in the most symmetrical curvilinear coordinate system and apply vector differential operators accurately.

3. **CLO 3 [Cognitive: Analyzing / Evaluating]:** Solve first- and second-order linear differential equations analytically, evaluating asymptotic behaviors, damping regimes, and resonance phenomena.

4. **CLO 4 [Practical / Skill: Creating / Computing]:** Implement scientific Python algorithms to compute numerical trajectories, evaluate matrix eigenvalues, and visualize multivariable vector fields.

Course Curriculum Map (9 Chapters)

Week	Chapter	Topic Overview	Target CLO
1–2	Chapter 1	Curvilinear Coordinate Systems & Scale Factors	CLO 1, 2
3–4	Chapter 2	Vector Analysis & Differential Field Operators	CLO 1, 2
5	Chapter 3	Infinite Series, Taylor Expansions & Approximations	CLO 1, 2
6–7	Chapter 4	Complex Analysis, Euler's Formula & Phasors	CLO 1, 3
8–9	Chapter 5	Matrices, Eigenvalues & Coupled Oscillations	CLO 2, 4
10	Chapter 6	Multivariable Differential Calculus & Optimization	CLO 2, 3
11–12	Chapter 7	Multiple Integrals & Integral Theorems (Gauss/Stokes)	CLO 2, 3
13	Chapter 8	First-Order Ordinary Differential Equations	CLO 3, 4
14–15	Chapter 9	Second-Order ODEs, Damped Oscillations & Resonance	CLO 3, 4

Table 0.1: Instructional schedule and curriculum alignment with CLOs.

Evaluation and Grading Policy

- **Continuous Assessment (Formative):** 60%
  - Homework Problem Sets (Tiered Conceptual and Analytical): 25%
  - Active Learning Python Simulation Mini-Projects: 20%
  - Midterm Examination (Chapters 1–5): 15%
- **Comprehensive Final Examination (Summative):** 40%

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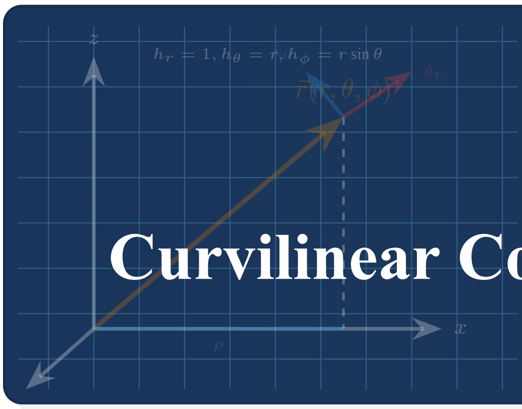


Chapter 1

Curvilinear Coordinate Systems & Scale Factors

Chapter 1 Instructional Management Plan

1. Chapter Topics and Core Content

1. Introduction and Cartesian Coordinate Foundations
2. Cylindrical Coordinate System
3. Spherical Coordinate System and GPS Geodesic Applications
4. Summary of Key Formulas and Coordinate Transformations
5. Chapter Exercises and Worked Physical Problems

2. Behavioral Learning Objectives (OBE)

1. Rigorously define and transform between Cartesian, cylindrical, and spherical coordinates.
2. Derive metric scale factors (Lamé coefficients h_i) for orthogonal curvilinear systems.
3. Formulate differential displacement, surface, and volume elements across coordinate bases.
4. Apply spherical coordinates to satellite geodesics and GPS positioning geometry.

3. Teaching Methods and Instructional Activities

1. Theoretical lecture integrated with 3D dynamic coordinate plane visualizations.
2. Collaborative problem-solving sessions on metric tensor and scale factor derivations.
3. Computational laboratory implementing coordinate transformations and plotting with Python.
4. Case study discussion on GPS satellite positioning and orbital geometry.

4. Instructional Media and Educational Technology

1. Course textbook: Chapter 1 Curvilinear Coordinate Systems & Scale Factors.
2. Electronic lecture slides with 3D interactive vector models.
3. Python scientific simulation scripts using NumPy and Matplotlib.
4. Digital learning management system and online computational exercises.

5. Measurement and Learning Assessment

1. Formative evaluation of in-class engagement and active problem-solving participation.
2. Assessment of homework problem sets on coordinate transformations and metric elements.
3. Chapter 1 diagnostic quiz evaluating analytical and conceptual mastery.

1.1 Introduction and Cartesian Coordinate Foundations

In theoretical physics, formulating dynamic laws, field distributions, and conservation principles requires a rigorous frame of reference. Selecting a coordinate system whose geometric symmetry reflects that of the physical boundary conditions significantly reduces the analytical complexity of governing differential equations.

The Cartesian coordinate system (x, y, z) represents the fundamental baseline. Three mutually orthogonal axes intersect at the origin $O(0, 0, 0)$, forming a right-handed basis characterized by fixed unit vectors $\hat{\imath}, \hat{\jmath}, \hat{k}$ satisfying the orthonormality condition:

$$\hat{e}_i \cdot \hat{e}_j = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases} \tag{1.1}$$

where δ_{ij} is the Kronecker delta. The position vector $\vec{r}$ of a point $P(x, y, z)$ is:

$$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k} \tag{1.2}$$

The differential displacement vector and squared infinitesimal arc length are:

$$d\vec{r} = dx\,\hat{\imath} + dy\,\hat{\jmath} + dz\,\hat{k} \tag{1.3}$$
$$ds^2 = |d\vec{r}|^2 = dx^2 + dy^2 + dz^2 \tag{1.4}$$

The differential volume element is simply $dV = dx\,dy\,dz$.

Definition 1.1 Orthogonal Curvilinear Coordinates and Scale Factors

Let (u_1, u_2, u_3) be generalized coordinates related to Cartesian coordinates by invertibly differentiable functions $x = x(u_1, u_2, u_3)$, $y = y(u_1, u_2, u_3)$, and $z = z(u_1, u_2, u_3)$. The system is **orthogonal** if the local basis vectors are mutually perpendicular. The **metric scale factors** (Lamé coefficients) are defined as:

$$h_i = \left|\frac{\partial \vec{r}}{\partial u_i}\right| = \sqrt{\left(\frac{\partial x}{\partial u_i}\right)^2 + \left(\frac{\partial y}{\partial u_i}\right)^2 + \left(\frac{\partial z}{\partial u_i}\right)^2} \tag{1.5}$$

The unit basis vectors are $\hat{e}_i = \frac{1}{h_i}\frac{\partial \vec{r}}{\partial u_i}$. The differential line and volume elements become:

$$ds^2 = h_1^2 du_1^2 + h_2^2 du_2^2 + h_3^2 du_3^2 \tag{1.6}$$
$$dV = h_1 h_2 h_3\, du_1\, du_2\, du_3 \tag{1.7}$$

For Cartesian coordinates, $h_x = h_y = h_z = 1$.

1.2 Cylindrical Coordinate System

Physical configurations exhibiting axial or rotational symmetry around a preferred axis—such as magnetic fields produced by long solenoids, laminar fluid flow through pipes, or rotating flywheels—are most naturally represented in cylindrical coordinates (ρ, ϕ, z).

Here, $\rho \in [0, \infty)$ is the perpendicular radial distance from the z-axis, $\phi \in [0, 2\pi)$ is the azimuthal angle measured counterclockwise from the positive x-axis, and $z \in (-\infty, \infty)$ is the axial coordinate:

$$x = \rho \cos\phi, \quad y = \rho \sin\phi, \quad z = z \tag{1.8}$$

Inverting these relations yields:

$$\rho = \sqrt{x^2 + y^2}, \quad \phi = \arctan\left(\frac{y}{x}\right), \quad z = z \tag{1.9}$$

To compute the scale factors, we differentiate $\vec{r}(\rho, \phi, z) = \rho \cos\phi\, \hat{\imath} + \rho \sin\phi\, \hat{\jmath} + z\, \hat{k}$:

$$\frac{\partial \vec{r}}{\partial \rho} = \cos\phi\, \hat{\imath} + \sin\phi\, \hat{\jmath} \implies h_\rho = 1 \tag{1.10}$$

$$\frac{\partial \vec{r}}{\partial \phi} = -\rho \sin\phi\, \hat{\imath} + \rho \cos\phi\, \hat{\jmath} \implies h_\phi = \rho \tag{1.11}$$

$$\frac{\partial \vec{r}}{\partial z} = \hat{k} \implies h_z = 1 \tag{1.12}$$

Thus, the orthonormal unit vectors are:

$$\hat{\rho} = \cos\phi\, \hat{\imath} + \sin\phi\, \hat{\jmath} \tag{1.13}$$

$$\hat{\phi} = -\sin\phi\, \hat{\imath} + \cos\phi\, \hat{\jmath} \tag{1.14}$$

$$\hat{k} = \hat{k} \tag{1.15}$$

The differential displacement, line element squared, and volume element are:

$$d\vec{r} = d\rho\, \hat{\rho} + \rho\, d\phi\, \hat{\phi} + dz\, \hat{k} \tag{1.16}$$

$$ds^2 = d\rho^2 + \rho^2 d\phi^2 + dz^2 \tag{1.17}$$

$$dV = \rho\, d\rho\, d\phi\, dz \tag{1.18}$$

Pitfall Alert: Time Derivatives of Non-Cartesian Unit Vectors

Unlike Cartesian unit vectors $(\hat{\imath}, \hat{\jmath}, \hat{k})$ which remain invariant in space and time, cylindrical and spherical unit vectors depend on position. When differentiating with respect to time t:

$$\dot{\hat{\rho}} = \frac{d\hat{\rho}}{dt} = \dot{\phi}\, \hat{\phi}, \quad \dot{\hat{\phi}} = -\dot{\phi}\, \hat{\rho} \tag{1.19}$$

Neglecting these geometric derivative terms produces severe errors in velocity and acceleration derivations.

1.3 Spherical Coordinate System

For central force fields (e.g., Newtonian gravity, Coulomb potential) and spherically symmetric wave propagation, spherical coordinates (r, θ, ϕ) are indispensable.

The coordinates are defined as:

- $r \in [0, \infty)$: radial distance from the origin.

- $\theta \in [0, \pi]$: polar (zenith) angle measured from the positive z-axis.
- $\phi \in [0, 2\pi)$: azimuthal angle in the xy-plane from the positive x-axis.

The Cartesian transformation equations are:

$$x = r \sin\theta \cos\phi, \quad y = r \sin\theta \sin\phi, \quad z = r \cos\theta \tag{1.20}$$

The metric scale factors are obtained from $\vec{r}(r, \theta, \phi)$:

$$h_r = \left|\frac{\partial \vec{r}}{\partial r}\right| = 1 \tag{1.21}$$

$$h_\theta = \left|\frac{\partial \vec{r}}{\partial \theta}\right| = r \tag{1.22}$$

$$h_\phi = \left|\frac{\partial \vec{r}}{\partial \phi}\right| = r \sin\theta \tag{1.23}$$

Consequently, the differential elements are:

$$d\vec{r} = dr\,\hat{r} + r\,d\theta\,\hat{\theta} + r\sin\theta\,d\phi\,\hat{\phi} \tag{1.24}$$

$$ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta\,d\phi^2 \tag{1.25}$$

$$dV = r^2 \sin\theta\,dr\,d\theta\,d\phi \tag{1.26}$$

Coordinate System	**Coordinates**	**Scale Factors** (h_1, h_2, h_3)	**Infinitesimal Volume** dV
Cartesian	(x, y, z)	$(1, 1, 1)$	$dx\,dy\,dz$
Cylindrical	(ρ, ϕ, z)	$(1, \rho, 1)$	$\rho\,d\rho\,d\phi\,dz$
Spherical	(r, θ, ϕ)	$(1, r, r\sin\theta)$	$r^2 \sin\theta\,dr\,d\theta\,d\phi$

Table 1.1: Comparison of metric scale factors and volume elements across standard coordinate systems.

1.4 Worked Examples and Physical Applications

Example 1.1: Kinematic Acceleration in Cylindrical Coordinates

A point particle moves in a plane ($z = 0$) such that its radial distance is $\rho(t) = \rho_0 e^{\alpha t}$ while rotating with a constant angular speed $\omega = \dot{\phi}$. Determine the general acceleration vector $\vec{a}(t)$ in cylindrical basis.

Picture / Conceptualization: The particle spirals outward in the xy-plane. The coordinate basis vectors $\hat{\rho}$ and $\hat{\phi}$ rotate with the particle at rate ω.

Strategy / Governing Laws: Express position $\vec{r} = \rho\hat{\rho}$. Differentiate twice with respect to time, applying the kinematic chain rule to account for $\dot{\hat{\rho}} = \dot{\phi}\hat{\phi}$ and $\dot{\hat{\phi}} = -\dot{\phi}\hat{\rho}$.

Calculation / Analytical Solution: 1. Velocity vector:

$$\vec{v} = \dot{\vec{r}} = \dot{\rho}\hat{\rho} + \rho\dot{\hat{\rho}} = \dot{\rho}\hat{\rho} + \rho\dot{\phi}\hat{\phi} \tag{1.27}$$

2. Acceleration vector:

$$\vec{a} = \frac{d\vec{v}}{dt} = \ddot{\rho}\hat{\rho} + \dot{\rho}\dot{\hat{\rho}} + (\dot{\rho}\dot{\phi} + \rho\ddot{\phi})\hat{\phi} + \rho\dot{\phi}\dot{\hat{\phi}}$$
$$= (\ddot{\rho} - \rho\dot{\phi}^2)\hat{\rho} + (2\dot{\rho}\dot{\phi} + \rho\ddot{\phi})\hat{\phi} \tag{1.28}$$

3. Substituting $\rho(t) = \rho_0 e^{\alpha t}$, $\dot{\rho} = \alpha\rho$, $\ddot{\rho} = \alpha^2\rho$, $\dot{\phi} = \omega$, and $\ddot{\phi} = 0$:

$$\vec{a}(t) = (\alpha^2 - \omega^2)\rho_0 e^{\alpha t}\hat{\rho} + 2\alpha\omega\rho_0 e^{\alpha t}\hat{\phi} \tag{1.29}$$

Think / Physical Insights: The radial component contains the centrifugal acceleration term $-\rho\omega^2$, while the azimuthal component $2\alpha\omega\rho$ is the Coriolis acceleration. When $\alpha = \omega$, the radial acceleration vanishes, yielding purely tangential acceleration. ■

Trajectory and Scale Factor Verification in Python

```python
import numpy as np
import matplotlib.pyplot as plt

# Parameters for spiral motion
t = np.linspace(0, 2, 500)
rho_0 = 1.0
alpha = 0.8
omega = 4.0

rho = rho_0 * np.exp(alpha * t)
phi = omega * t

x = rho * np.cos(phi)
y = rho * np.sin(phi)

# Acceleration components
a_rho = (alpha**2 - omega**2) * rho
a_phi = 2 * alpha * omega * rho

print(f"At t = 1.0s: rho = {rho_0*np.exp(alpha):.3f} m")
print(f"Radial acceleration: {a_rho[250]:.3f} m/s^2, Tangential: {a_phi
    [250]:.3f} m/s^2")
```

Example 1.2: GPS Geodesic Distance from Spherical Coordinates

Two points on Earth's surface are given by geographic coordinates: Bangkok ($\lambda_1 = 13.75°$N, $\phi_1 = 100.52°$E) and Tokyo ($\lambda_2 = 35.68°$N, $\phi_2 = 139.65°$E). Using the spherical law of cosines, calculate the great-circle (geodesic) distance between them. Take Earth's mean radius $R_\oplus = 6371$ km.

Solution:

Step 1 — Convert to spherical coordinates. In physics conventions, the polar angle θ is measured from the North Pole, so $\theta = 90° - \lambda$:

$$\theta_1 = 90° - 13.75° = 76.25°, \quad \phi_1 = 100.52° \tag{1.30}$$

$$\theta_2 = 90° - 35.68° = 54.32°, \quad \phi_2 = 139.65° \tag{1.31}$$

Step 2 — Apply the spherical law of cosines. The angle $\Delta\sigma$ subtended at Earth's centre between two points on a unit sphere satisfies:

$$\cos(\Delta\sigma) = \cos\theta_1 \cos\theta_2 + \sin\theta_1 \sin\theta_2 \cos(\phi_2 - \phi_1) \tag{1.32}$$

Substituting $\Delta\phi = 139.65° - 100.52° = 39.13°$:

$$\cos(\Delta\sigma) = \cos(76.25°)\cos(54.32°) + \sin(76.25°)\sin(54.32°)\cos(39.13°) \tag{1.33}$$

$$= (0.2385)(0.5835) + (0.9712)(0.8122)(0.7753) \tag{1.34}$$

$$= 0.1391 + 0.6124 = 0.7515 \tag{1.35}$$

Therefore $\Delta\sigma = \arccos(0.7515) = 41.27° = 0.7203$ rad.

Step 3 — Compute the geodesic distance:

$$\boxed{d = R_\oplus \cdot \Delta\sigma = 6371 \times 0.7203 \approx 4589 \text{ km}} \tag{1.36}$$

Think / Physical Insights: This calculation underlies all GPS receiver algorithms. The Haversine formula (used in navigation software) is numerically superior to the simple cosine formula when $\Delta\sigma \to 0$, because it avoids catastrophic cancellation. For intercontinental distances, however, both formulas agree to better than 0.1%. ■

Great-Circle Distance Verification in Python

```python
import numpy as np

def great_circle(lat1, lon1, lat2, lon2, R=6371):
    # Convert degrees to radians
    t1, t2 = np.radians(90 - lat1), np.radians(90 - lat2)
    p1, p2 = np.radians(lon1), np.radians(lon2)
    cos_dsigma = np.cos(t1)*np.cos(t2) + np.sin(t1)*np.sin(t2)*np.cos(
    p2 - p1)
    dsigma = np.arccos(np.clip(cos_dsigma, -1, 1))
    return R * dsigma

d = great_circle(13.75, 100.52, 35.68, 139.65)
print(f"Bangkok → Tokyo great-circle distance: {d:.1f} km")
print(f"Subtended angle: {np.degrees(d/6371):.2f} degrees")
```

1.5 Summary of Key Formulas

Concept	**Formula**	**Physical Significance**
Metric Scale Factor	$h_i = \lvert\partial\vec{r}/\partial u_i\rvert$	Local arc length per unit coordinate variation
Infinitesimal Line Element	$ds^2 = \sum_i h_i^2 du_i^2$	Invariant metric distance in differential geometry
Cylindrical Basis Derivatives	$\dot{\hat{\rho}} = \dot{\phi}\hat{\phi},\ \dot{\hat{\phi}} = -\dot{\phi}\hat{\rho}$	Origin of centripetal and Coriolis inertial terms
Spherical Solid Angle	$d\Omega = \sin\theta\, d\theta\, d\phi$	Measure of angular aperture in radiation and scattering
Great-Circle Distance	$d = R \arccos(\cos\theta_1 \cos\theta_2 + \sin\theta_1 \sin\theta_2 \cos\Delta\phi)$	Shortest path on a sphere (GPS, aviation)

Table 1.2: Summary of coordinate geometry relations and physical interpretations.

1.6 Chapter Exercises

1. **Analytical Transformation:** Find the spherical coordinates (r, θ, ϕ) of the point given in Cartesian coordinates by $P(-2, 2\sqrt{3}, -4)$.

2. **Scale Factor Derivation:** Consider parabolic cylindrical coordinates (u, v, z) defined by $x = \frac{1}{2}(u^2 - v^2)$, $y = uv$, $z = z$. Calculate scale factors h_u, h_v, h_z and show that the coordinate system is orthogonal.

3. **Solid Angle of a Cone:** A right-circular cone has a semi-vertical angle α. By integrating the spherical solid angle $d\Omega = \sin\theta\, d\theta\, d\phi$, prove that the solid angle subtended at the apex is $\Omega = 2\pi(1 - \cos\alpha)$.

4. **Velocity in Spherical Coordinates:** A particle moves on a sphere of radius R with $\dot{\theta} = \omega_\theta$ (const) and $\dot{\phi} = \omega_\phi$ (const). Write the velocity vector $\vec{v}$ in the spherical basis $(\hat{r}, \hat{\theta}, \hat{\phi})$ and compute its magnitude. Under what condition is $|\vec{v}|$ constant?

5. **Ellipsoidal Coordinates:** Prolate spheroidal coordinates (ξ, η, ϕ) are related to Cartesian coordinates by $x = a \sinh \xi \sin \eta \cos \phi$, $y = a \sinh \xi \sin \eta \sin \phi$, $z = a \cosh \xi \cos \eta$ where $a > 0$. Derive the three scale factors and the volume element dV.

Chapter 2

Vector Analysis & Differential Field Operators

Chapter 2 Instructional Management Plan

1. Chapter Topics and Core Content

1. Vector Algebra and Triple Products
2. Differential Field Operators: Gradient, Divergence, and Curl
3. Conservative Vector Fields and Potential Theory
4. Worked Examples and Electrodynamic Applications
5. Summary of Key Formulas and Chapter Exercises

2. Behavioral Learning Objectives (OBE)

1. Perform algebraic operations with dot, cross, and scalar/vector triple products.
2. Calculate and physically interpret gradient (∇), divergence ($\nabla\cdot$), and curl ($\nabla\times$).
3. Construct scalar potentials for conservative fields and verify solenoidal conditions.
4. Apply vector differential identities to electromagnetic fields and fluid dynamics.

3. Teaching Methods and Instructional Activities

1. Interactive lecture analyzing spatial gradients, flux divergence, and circulation curl.
2. Analytical workshops deriving fundamental vector differential identities.
3. Computational simulation lab visualizing 2D/3D vector fields and streamlines with Python.
4. Problem-based learning applying potential theory to electrostatic and gravitational fields.

4. Instructional Media and Educational Technology

1. Course textbook: Chapter 2 Vector Analysis & Differential Field Operators.
2. Multimedia lecture slides featuring field contour maps and curl vortex animations.
3. Python scripts for vector field quiver plots using Matplotlib and NumPy.
4. Online problem sets and interactive vector algebra modules.

5. Measurement and Learning Assessment

1. Formative assessment through classroom response questions and vector identity proofs.
2. Evaluation of weekly homework problem sets on field operators and potential theory.
3. Chapter 2 diagnostic quiz covering analytical calculations and physical field interpretations.

2.1 Vector Algebra and Triple Products

Physical quantities characterized by both magnitude and spatial orientation are represented mathematically as vectors. In three-dimensional Euclidean space, let $\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}$ and $\vec{B} = B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}$.

The scalar (dot) product is defined by:

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos\theta = A_xB_x + A_yB_y + A_zB_z \tag{2.1}$$

The vector (cross) product produces an orthogonal axial vector according to the right-hand rule:

$$\vec{A} \times \vec{B} = |\vec{A}||\vec{B}| \sin\theta\, \hat{n} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \tag{2.2}$$

2.1.1 Triple Products and Geometric Interpretations

The **scalar triple product** represents the signed volume of the parallelepiped subtended by vectors $\vec{A}, \vec{B}, \vec{C}$:

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} \tag{2.3}$$

This product is cyclic: $\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B})$.

The **vector triple product** satisfies the celebrated "BAC–CAB" identity:

$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B}) \tag{2.4}$$

This identity is fundamental in classical mechanics for formulating the angular momentum of rotating rigid bodies $\vec{L} = \sum m_i[\vec{r}_i \times (\vec{\omega} \times \vec{r}_i)]$.

2.2 Differential Field Operators: Gradient, Divergence, and Curl

Spatial variation of physical fields requires the Del (Nabla) vector differential operator:

$$\nabla \equiv \hat{\imath}\frac{\partial}{\partial x} + \hat{\jmath}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z} \tag{2.5}$$

Definition 2.1 The Gradient of a Scalar Field

For a scalar field $\psi(x, y, z)$, the **gradient** $\nabla\psi$ is a vector field pointing in the direction of the greatest rate of spatial increase of ψ, with magnitude equal to that maximum directional rate of change:

$$\nabla\psi = \frac{\partial\psi}{\partial x}\hat{\imath} + \frac{\partial\psi}{\partial y}\hat{\jmath} + \frac{\partial\psi}{\partial z}\hat{k} \tag{2.6}$$

Geometrically, $\nabla\psi$ is perpendicular to the equipotential or level surface $\psi(x, y, z) = C$ at every point.

Definition 2.2 The Divergence of a Vector Field

The **divergence** of a vector field $\vec{A}$ represents the net volumetric flux density outflow per unit infinitesimal volume from a given point:

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \tag{2.7}$$

If $\nabla \cdot \vec{A} = 0$ everywhere, the field is called **solenoidal** (e.g., the magnetic field $\vec{B}$ in Maxwell's second law $\nabla \cdot \vec{B} = 0$, indicating the non-existence of magnetic monopoles).

Definition 2.3 The Curl of a Vector Field

The **curl** of a vector field $\vec{A}$ measures the microscopic circulation density or rotational vorticity around an infinitesimal loop:

$$\nabla \times \vec{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{i} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{j} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{k} \tag{2.8}$$

A vector field with $\nabla \times \vec{A} = \vec{0}$ is called **irrotational** or **conservative**.

Theorem 2.1 Null Identities of Vector Calculus

For any twice continuously differentiable scalar field ψ and vector field $\vec{A}$:

$$\nabla \times (\nabla\psi) = \vec{0} \quad \text{(The curl of any gradient field is identically zero)} \tag{2.9}$$

$$\nabla \cdot (\nabla \times \vec{A}) = 0 \quad \text{(The divergence of any curl field is identically zero)} \tag{2.10}$$

Pitfall Alert: Common Operator Confusion

Do not confuse the Laplacian of a scalar with the vector Laplacian. For a scalar field:

$$\nabla^2\psi = \nabla \cdot (\nabla\psi) = \frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} \tag{2.11}$$

For a vector field, the vector Laplacian is defined via the curl-of-curl identity:

$$\nabla^2\vec{A} \equiv \nabla(\nabla \cdot \vec{A}) - \nabla \times (\nabla \times \vec{A}) \tag{2.12}$$

In Cartesian coordinates, $\nabla^2\vec{A} = (\nabla^2 A_x)\hat{i} + (\nabla^2 A_y)\hat{j} + (\nabla^2 A_z)\hat{k}$, but this component-wise equality fails in curvilinear coordinates because unit vectors depend on spatial coordinates!

2.3 Conservative Vector Fields and Potential Theory

A vector force field $\vec{F}$ is conservative if the work done $W = \int_C \vec{F} \cdot d\vec{r}$ between any two points is independent of the path. The following four statements are mathematically equivalent in a simply connected domain:

1. $\oint_C \vec{F} \cdot d\vec{r} = 0$ for every closed contour C.
2. $\int_A^B \vec{F} \cdot d\vec{r}$ is path-independent.
3. $\vec{F}$ can be expressed as the negative gradient of a scalar potential: $\vec{F} = -\nabla V$.
4. $\vec{F}$ is irrotational everywhere: $\nabla \times \vec{F} = \vec{0}$.

2.4 Worked Examples and Physical Applications

Example 2.1: Testing Conservatism and Finding the Potential Function

Consider the gravitational/electrostatic type force field:

$$\vec{F}(x,y,z) = (2xy + z^3)\hat{\imath} + x^2\hat{\jmath} + 3xz^2\hat{k} \tag{2.13}$$

Determine whether $\vec{F}$ is conservative. If so, construct its potential energy function $V(x,y,z)$ such that $\vec{F} = -\nabla V$ with $V(0,0,0) = 0$.

Picture / Conceptualization: The field components depend on polynomial combinations of spatial coordinates. We test whether the field possesses microscopic vorticity.

Strategy / Governing Laws: Compute $\nabla \times \vec{F}$. If the curl vanishes identically, determine V by integrating the partial derivative conditions $-\frac{\partial V}{\partial x} = F_x$, $-\frac{\partial V}{\partial y} = F_y$, $-\frac{\partial V}{\partial z} = F_z$.

Calculation / Analytical Solution: 1. Evaluate the curl:

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + z^3 & x^2 & 3xz^2 \end{vmatrix} = \hat{\imath}(0-0) - \hat{\jmath}(3z^2 - 3z^2) + \hat{k}(2x - 2x) = \vec{0} \tag{2.14}$$

Because $\nabla \times \vec{F} = \vec{0}$ everywhere, $\vec{F}$ is conservative.

2. Integrate for $V(x,y,z)$:

$$\frac{\partial V}{\partial x} = -(2xy + z^3) \implies V(x,y,z) = -x^2y - xz^3 + g(y,z) \tag{2.15}$$

$$\frac{\partial V}{\partial y} = -x^2 + \frac{\partial g}{\partial y} = -x^2 \implies \frac{\partial g}{\partial y} = 0 \implies g(y,z) = h(z) \tag{2.16}$$

$$\frac{\partial V}{\partial z} = -3xz^2 + h'(z) = -3xz^2 \implies h'(z) = 0 \implies h(z) = C \tag{2.17}$$

Given the reference condition $V(0,0,0) = 0$, the integration constant is $C = 0$. Thus:

$$V(x,y,z) = -(x^2y + xz^3) \tag{2.18}$$

Think / Physical Insights: Because $\vec{F} = -\nabla V$, moving a test particle along any closed trajectory in this field yields net zero energy dissipation, ensuring complete conservation of mechanical energy $E = T + V$. ■

Vector Field Divergence and Streamline Visualization

```python
import numpy as np
import matplotlib.pyplot as plt

# Define 2D grid
Y, X = np.mgrid[-2:2:100j, -2:2:100j]

# Dipole-like electric vector field
U = X / (X**2 + Y**2 + 0.1)**1.5
V = Y / (X**2 + Y**2 + 0.1)**1.5

```

```
# Numerical divergence: dU/dx + dV/dy
dx = X[0, 1] - X[0, 0]
dy = Y[1, 0] - Y[0, 0]
div = np.gradient(U, axis=1)/dx + np.gradient(V, axis=0)/dy

print(f"Max divergence near core: {np.max(div):.2f}")
print("Streamlines smoothly diverge radially outwards from the positive
    source.")
```

Example 2.2: Magnetic Torque on a Current Loop

A rectangular current loop of width a and height b carries a steady current I. The loop lies in the xy-plane with its normal $\hat{n} = \hat{k}$. An external uniform magnetic field $\vec{B} = B_0(\hat{\imath} + \sqrt{3}\,\hat{\jmath})$ is applied. Find (i) the net magnetic force on the loop, (ii) the magnetic torque $\vec{\tau} = \vec{m} \times \vec{B}$, and (iii) the potential energy $U = -\vec{m} \cdot \vec{B}$.

Solution:

The magnetic dipole moment is $\vec{m} = I(ab)\hat{k} = IAb\hat{k}$ where $A = ab$.

(i) Net force on a rigid current loop in a uniform field is zero:

$$\vec{F}_{\text{net}} = I \oint (d\vec{l} \times \vec{B}) = I \left(\oint d\vec{l} \right) \times \vec{B} = \vec{0} \tag{2.19}$$

since $\oint d\vec{l} = \vec{0}$ for a closed path.

(ii) Magnetic torque:

$$\vec{\tau} = \vec{m} \times \vec{B} = (IAb\hat{k}) \times B_0(\hat{\imath} + \sqrt{3}\,\hat{\jmath}) \tag{2.20}$$

$$= IAbB_0 \left[(\hat{k} \times \hat{\imath}) + \sqrt{3}(\hat{k} \times \hat{\jmath}) \right] \tag{2.21}$$

$$= IAbB_0 \left[\hat{\jmath} + \sqrt{3}(-\hat{\imath}) \right] = IAbB_0(-\sqrt{3}\,\hat{\imath} + \hat{\jmath}) \tag{2.22}$$

The torque magnitude is:

$$\boxed{|\vec{\tau}| = IAbB_0\sqrt{3+1} = 2IAbB_0} \tag{2.23}$$

(iii) Potential energy:

$$U = -\vec{m} \cdot \vec{B} = -(IAb\hat{k}) \cdot B_0(\hat{\imath} + \sqrt{3}\,\hat{\jmath}) = 0 \tag{2.24}$$

The potential energy is zero because $\vec{m} \perp \vec{B}$ (the loop is in a metastable orientation).

Think / Physical Insights: The torque $\vec{\tau}$ acts to align $\vec{m}$ with $\vec{B}$, reducing U to its minimum $-|\vec{m}||\vec{B}|$. This is the fundamental operating principle of electric motors and galvanometers. The torque is maximum when $\vec{m} \perp \vec{B}$ (as here) and vanishes at equilibrium. ■

2.5 Summary of Key Formulas

Operator / Law	Mathematical Definition	Physical Meaning
Gradient $\nabla\psi$	$\sum_i \frac{1}{h_i}\frac{\partial\psi}{\partial u_i}\hat{e}_i$	Maximum directional spatial derivative; normal to level surfaces
Divergence $\nabla\cdot\vec{A}$	$\frac{1}{h_1h_2h_3}\sum_i \frac{\partial}{\partial u_i}\left(\frac{h_1h_2h_3}{h_i}A_i\right)$	Net outflow flux density per unit volume (sources and sinks)
Curl $\nabla\times\vec{A}$	$\frac{1}{h_1h_2h_3}\begin{vmatrix} h_1\hat{e}_1 & h_2\hat{e}_2 & h_3\hat{e}_3 \\ \partial/\partial u_1 & \partial/\partial u_2 & \partial/\partial u_3 \\ h_1A_1 & h_2A_2 & h_3A_3 \end{vmatrix}$	Circulation vorticity around microscopic loops
Laplacian $\nabla^2\psi$	$\nabla\cdot(\nabla\psi)$	Mean curvature; appears in Poisson and wave equations
Magnetic Torque	$\vec{\tau}=\vec{m}\times\vec{B},\ U=-\vec{m}\cdot\vec{B}$	Aligns current loop with external field (motor principle)

Table 2.1: Summary of differential vector operators in orthogonal curvilinear coordinates.

2.6 Chapter Exercises

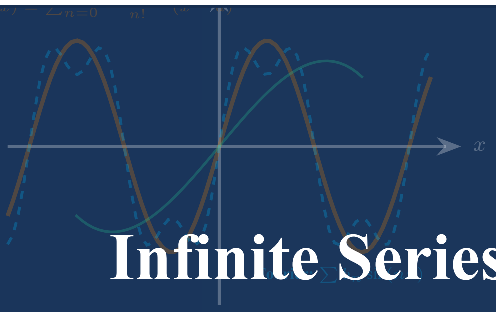
1. **BAC-CAB Expansion:** Prove algebraically that $(\vec{A}\times\vec{B})\cdot(\vec{C}\times\vec{D}) = (\vec{A}\cdot\vec{C})(\vec{B}\cdot\vec{D}) - (\vec{A}\cdot\vec{D})(\vec{B}\cdot\vec{C})$.

2. **Angular Velocity Field:** The velocity field of a rigid body rotating with constant angular velocity $\vec{\omega} = \omega\hat{k}$ is $\vec{v} = \vec{\omega}\times\vec{r}$. Calculate $\nabla\cdot\vec{v}$ and $\nabla\times\vec{v}$. What is the physical significance of $\nabla\times\vec{v}$?

3. **Solenoidal and Irrotational Check:** Show that the field $\vec{B}(\vec{r}) = \frac{\vec{m}\times\vec{r}}{r^3}$ (magnetic dipole field) satisfies $\nabla\cdot\vec{B} = 0$ for $r \neq 0$.

4. **Orbital Angular Momentum:** An electron in a hydrogen atom moves in a circular orbit of radius $a_0 = 0.529$ Å with speed $v = 2.19\times10^6$ m/s. Calculate the orbital angular momentum $\vec{L} = m_e\vec{r}\times\vec{v}$ and show that $|\vec{L}| = \hbar$ (the reduced Planck constant).

5. **Stokes' Theorem Application:** The vector field $\vec{F} = (y^2-z^2)\hat{\imath} + (z^2-x^2)\hat{\jmath} + (x^2-y^2)\hat{k}$. Use Stokes' theorem to evaluate $\oint_C \vec{F}\cdot d\vec{l}$ where C is the intersection of the sphere $x^2+y^2+z^2=1$ with the plane $x+y+z=0$.



Chapter 3

Infinite Series & Approximations

Chapter 3 Instructional Management Plan

1. Chapter Topics and Core Content

1. Introduction and Power Series Representation
2. Convergence Criteria and Radius of Convergence
3. Physical Applications: Asymptotic Expansions
4. Worked Examples and Relativistic Energy Approximations
5. Summary of Key Formulas and Chapter Exercises

2. Behavioral Learning Objectives (OBE)

1. Expand functions into Taylor and Maclaurin power series systematically.
2. Determine radii and intervals of convergence using analytical tests.
3. Apply binomial expansions to relativistic kinetic energy and multipole fields.
4. Quantify truncation errors and higher-order asymptotic corrections.

3. Teaching Methods and Instructional Activities

1. Lecture on power series foundations and convergence criteria.
2. Workshop calculating series expansions for physical potentials.
3. Python computational lab analyzing series truncation error convergence.
4. Guided group analysis of relativistic limits and pendulum approximations.

4. Instructional Media and Educational Technology

1. Course textbook: Chapter 3 Infinite Series & Approximations.
2. Slide presentations displaying partial sum convergence animations.
3. Python scripts for symbolic series expansion using SymPy.
4. Practice worksheets and homework problem sets.

5. Measurement and Learning Assessment

1. Evaluation of active class participation and convergence proof exercises.
2. Grading of problem sets on Taylor series and physical limits.
3. Chapter 3 diagnostic quiz on series manipulation and asymptotic approximations.

3.1 Introduction and Power Series Representation

Many fundamental differential equations in physics cannot be integrated into elementary closed-form expressions. Infinite series expansions serve as one of the most powerful analytical instruments in mathematical physics, enabling us to approximate complex potentials, linearize nonlinear dynamical equations, and study asymptotic limits.

A general power series centered at $x = x_0$ is defined as:

$$S(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \cdots \tag{3.1}$$

Theorem 3.1 Taylor and Maclaurin Series Expansion

If a real- or complex-valued function $f(x)$ is infinitely differentiable (C^∞) in an open neighborhood containing x_0, it may be represented by its **Taylor series**:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n \tag{3.2}$$

When the expansion is centered at the origin ($x_0 = 0$), it is designated as a **Maclaurin series**:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots \tag{3.3}$$

3.2 Convergence Criteria and Radius of Convergence

A power series is physically meaningful only within its domain of convergence. The series $\sum a_n (x - x_0)^n$ converges absolutely inside the interval $|x - x_0| < R$, where R is the **radius of convergence**.

Definition 3.1 Cauchy-Hadamard and D'Alembert Ratio Tests

The radius of convergence R can be calculated directly from the series coefficients:

$$\frac{1}{R} = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| \quad \text{or} \quad \frac{1}{R} = \lim_{n\to\infty} \sqrt[n]{|a_n|} \tag{3.4}$$

- If $R = \infty$, the series converges for all $x \in \mathbb{R}$ (e.g., $e^x, \sin x, \cos x$).
- If $0 < R < \infty$, the series converges for $|x - x_0| < R$ and diverges for $|x - x_0| > R$.
- The boundary points $x = x_0 \pm R$ must be investigated individually using the Alternating Series (Leibniz) test or the Integral test.

Function	Maclaurin Series Expansion	Radius R	Interval
e^x	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$	∞	$(-\infty, \infty)$
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$	∞	$(-\infty, \infty)$
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$	∞	$(-\infty, \infty)$
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}$	1	$(-1, 1]$
$(1+x)^p$	$1 + px + \frac{p(p-1)}{2!}x^2 + \frac{p(p-1)(p-2)}{3!}x^3 + \cdots$	1	$(-1, 1)$

Table 3.1: Standard Maclaurin expansions of elementary mathematical physics functions.

Pitfall Alert: Truncation Outside the Radius of Convergence

Approximating a function by the first few terms of its Taylor series is only valid when $|x - x_0| \ll R$. Attempting to compute $(1+x)^{-1/2}$ for $x = 2.5$ using $1 - \frac{1}{2}x + \frac{3}{8}x^2$ leads to diverging oscillations, because x lies outside the unit convergence radius $R = 1$.

3.3 Physical Applications: Asymptotic Expansions

3.3.1 Relativistic Kinetic Energy

In special relativity, the total energy of a particle with rest mass m moving at speed v is $E = \gamma mc^2$, where $\gamma = (1 - v^2/c^2)^{-1/2}$. The relativistic kinetic energy is:

$$K = E - mc^2 = (\gamma - 1)mc^2 = \left[\left(1 - \frac{v^2}{c^2}\right)^{-1/2} - 1\right] mc^2 \tag{3.5}$$

In the non-relativistic limit ($v \ll c$, so $\beta \equiv v/c \ll 1$), we apply the generalized binomial expansion $(1 - \beta^2)^{-1/2} = 1 + \frac{1}{2}\beta^2 + \frac{3}{8}\beta^4 + \frac{5}{16}\beta^6 + \mathcal{O}(\beta^8)$:

$$K = \left(1 + \frac{1}{2}\frac{v^2}{c^2} + \frac{3}{8}\frac{v^4}{c^4} + \cdots - 1\right) mc^2 = \frac{1}{2}mv^2 + \frac{3}{8}m\frac{v^4}{c^2} + \mathcal{O}\left(\frac{v^6}{c^4}\right) \tag{3.6}$$

The leading term recovers classical Newtonian kinetic energy $K_{\text{classical}} = \frac{1}{2}mv^2$, while the second term provides the first-order relativistic correction!

3.4 Worked Examples and Physical Applications

Example 3.1: Large-Amplitude Simple Pendulum Period Correction

The exact differential equation of an undamped simple pendulum of length L is:

$$\frac{d^2\theta}{dt^2} + \frac{g}{L}\sin\theta = 0 \tag{3.7}$$

Using the Maclaurin expansion of $\sin\theta$, obtain the third-order governing equation and derive the fractional correction to the natural frequency $\omega_0 = \sqrt{g/L}$ for finite amplitude θ_0.

Picture / Conceptualization: For tiny amplitudes $\theta \ll 1$, $\sin\theta \approx \theta$, giving simple harmonic motion. For larger amplitudes, restoring torque is weaker than linear, lengthening the oscillation period.
Strategy / Governing Laws: Expand $\sin\theta = \theta - \frac{\theta^3}{6} + \mathcal{O}(\theta^5)$. Substitute this into the equation of motion to produce the Duffing-type nonlinear oscillator:

$$\ddot{\theta} + \omega_0^2\left(\theta - \frac{\theta^3}{6}\right) = 0 \tag{3.8}$$

Calculation / Analytical Solution: Assume an approximate harmonic solution $\theta(t) \approx \theta_0\cos(\omega t)$. Using the trigonometric identity $\cos^3(\omega t) = \frac{3}{4}\cos(\omega t) + \frac{1}{4}\cos(3\omega t)$, substitute into the nonlinear term:

$$-\omega^2\theta_0\cos(\omega t) + \omega_0^2\left[\theta_0\cos(\omega t) - \frac{\theta_0^3}{6}\left(\frac{3}{4}\cos(\omega t) + \frac{1}{4}\cos(3\omega t)\right)\right] = 0 \tag{3.9}$$

Equating the secular fundamental harmonic coefficients of $\cos(\omega t)$:

$$-\omega^2 + \omega_0^2\left(1 - \frac{\theta_0^2}{8}\right) = 0 \implies \omega \approx \omega_0\sqrt{1 - \frac{\theta_0^2}{8}} \approx \omega_0\left(1 - \frac{\theta_0^2}{16}\right) \tag{3.10}$$

The modified period $T = \frac{2\pi}{\omega}$ becomes:

$$T \approx \frac{T_0}{1 - \frac{\theta_0^2}{16}} \approx T_0\left(1 + \frac{\theta_0^2}{16}\right) \tag{3.11}$$

Think / Physical Insights: For an initial amplitude $\theta_0 = 30^\circ \approx 0.524$ rad, the period increases by approximately $\frac{(0.524)^2}{16} \approx 1.7\%$. This demonstrates why high-precision mechanical clocks require amplitude stabilization. ■

Taylor Series Truncation Error Analysis in Python

```python
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-2, 2, 200)
f_exact = np.sin(x)

# Taylor polynomials centered at 0
p1 = x
p3 = x - x**3 / 6.0
p5 = x - x**3 / 6.0 + x**5 / 120.0

err_p1 = np.abs(f_exact - p1)
err_p3 = np.abs(f_exact - p3)
err_p5 = np.abs(f_exact - p5)

print(f"At x = 1.0 rad: Exact={np.sin(1.0):.6f}")
print(f"P1 error: {err_p1[150]:.6f}, P3 error: {err_p3[150]:.6f}, P5
    error: {err_p5[150]:.6f}")
```

Example 3.2: Fourier Series of a Square Wave

A periodic square wave with period $T = 2\pi$ is defined by:

$$f(x) = \begin{cases} +1, & 0 < x < \pi \\ -1, & \pi < x < 2\pi \end{cases}$$

Derive the Fourier series and calculate the partial sums S_1, S_3, and S_5. Verify the Gibbs phenomenon.
Solution:
The Fourier coefficients are:

$$a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)\,dx = 0 \quad \text{(odd function)} \tag{3.12}$$

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos(nx)\,dx = 0 \quad \text{(odd integrand)} \tag{3.13}$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin(nx)\,dx = \frac{2}{n\pi}[1 - \cos(n\pi)] \tag{3.14}$$

Since $1 - \cos(n\pi) = 0$ for even n and $= 2$ for odd n:

$$\boxed{f(x) = \frac{4}{\pi}\sum_{k=0}^{\infty} \frac{\sin\big((2k+1)x\big)}{2k+1} = \frac{4}{\pi}\left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots\right)} \tag{3.15}$$

At $x = \pi/2$, the series gives the Leibniz formula: $f(\pi/2) = \frac{4}{\pi}(1 - \frac{1}{3} + \frac{1}{5} - \cdots) = 1$ ✓

Think / Physical Insights: The Gibbs phenomenon is the $\approx 9\%$ overshoot (1.0895 instead of 1) near the discontinuity that persists no matter how many harmonics are included. It originates from the inability of any finite set of smooth sinusoids to represent a sharp jump discontinuity. In signal processing, the Gibbs phenomenon motivates windowed Fourier transforms (Hann, Hamming windows). ■

Fourier Series Partial Sums and Gibbs Phenomenon in Python

```python
import numpy as np

x = np.linspace(0.01, 2*np.pi - 0.01, 2000)

def fourier_square(x, N):
    s = np.zeros_like(x)
    for k in range(N):
        n = 2*k + 1
        s += np.sin(n * x) / n
    return 4/np.pi * s

for N in [1, 3, 5, 21]:
    s = fourier_square(x, N)
    print(f"N={N:2d} harmonics: max value = {s.max():.4f}  (Gibbs overshoot: {100*(s.max()-1):.1f}%)")
```

3.5 Summary of Key Formulas

Expansion / Test	Mathematical Formula	Physical Application
Taylor Series	$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$	Perturbation theory, equilibrium potential expansions
Ratio Test Radius	$R = \lim_{n\to\infty} \lvert a_n/a_{n+1} \rvert$	Valid boundary limits for mathematical physics models
Binomial Expansion	$(1 + x)^p = 1 + px + \frac{p(p-1)}{2}x^2 + \cdots$	Relativistic velocity limits, electrostatic multipole fields
Small Angle Approximation	$\cos\theta \approx 1 - \frac{\theta^2}{2}$, $\sin\theta \approx \theta$	Harmonic oscillator approximations in mechanics and optics
Fourier Series (odd)	$f(x) = \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\sin((2k+1)x)}{2k+1}$	Signal decomposition, heat conduction, quantum wave packets

Table 3.2: Summary of infinite series expansions and physical approximations.

3.6 Chapter Exercises

1. **Electric Dipole Potential:** Two point charges $+q$ and $-q$ are situated on the z-axis at $z = +d/2$ and $z = -d/2$. Write the exact potential at $(0, 0, z)$ for $z > d/2$, and use the binomial series to show that the leading term is the dipole potential $V \approx \frac{qd}{4\pi\epsilon_0 z^2}$.

2. **Planck's Radiation Law Limit:** Planck's spectral energy density is given by $\rho(\nu) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1}$. Expand the exponential for $h\nu \ll k_B T$ to deduce the classical Rayleigh-Jeans formula.

3. **Radius of Convergence:** Determine the radius of convergence of the series $\sum_{n=1}^{\infty} \frac{n^2 x^n}{3^n (n+1)!}$.

4. **Fourier Coefficient Parseval's Theorem:** For a square-integrable function $f(x)$ on $[-\pi, \pi]$ with Fourier coefficients a_n, b_n, Parseval's theorem states $\frac{1}{\pi} \int_{-\pi}^{\pi} |f(x)|^2 \, dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$. Apply Parseval's theorem to the square wave above to prove $\sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}$.

5. **Asymptotic Expansion of Stirling:** The Gamma function satisfies $\ln \Gamma(n + 1) = n \ln n - n + \frac{1}{2} \ln(2\pi n) + O(1/n)$. Use this to estimate the number of digits in 100! (i.e., $\lfloor \log_{10}(100!) \rfloor + 1$).

Chapter 4

Complex Analysis & Applications

Chapter 4 Instructional Management Plan

1. Chapter Topics and Core Content

1. Algebraic and Geometric Foundations of Complex Numbers
2. Euler's Formula, Polar Form, and Roots of Unity
3. Physical Applications: Phasors and AC Impedance
4. Worked Examples and Harmonic Wave Interference
5. Summary of Key Formulas and Chapter Exercises

2. Behavioral Learning Objectives (OBE)

1. Represent complex numbers interchangeably in Cartesian, polar, and exponential forms.
2. Apply de Moivre's theorem to compute roots of unity and trigonometric identities.
3. Model AC circuits and damped oscillations using complex impedance and phasors.
4. Verify analyticity of complex functions using Cauchy-Riemann equations.

3. Teaching Methods and Instructional Activities

1. Lecture on the Argand plane, Euler's formula, and complex exponential mappings.
2. Hands-on workshop deriving AC circuit impedance and phasor addition.
3. Computational lab simulating phasor rotations and frequency responses in Python.
4. Case study on resonance in RLC circuits and optical wave superposition.

4. Instructional Media and Educational Technology

1. Course textbook: Chapter 4 Complex Analysis & Applications.
2. Interactive digital slides displaying rotating phasor diagrams.
3. Python scripts for Bode plots and complex impedance loci.
4. Problem sets on complex algebra and AC circuit analysis.

5. Measurement and Learning Assessment

1. Evaluation of class participation and phasor diagram sketch exercises.
2. Grading of homework problem sets on complex impedance and Euler's formula.
3. Chapter 4 diagnostic quiz testing complex algebra and circuit analysis.

4.1 Algebraic and Geometric Foundations of Complex Numbers

In classical mechanics, optics, and quantum mechanics, physical quantities that oscillate with specific phase relationships are most elegantly described using complex variables. A complex number $z \in \mathbb{C}$ is an ordered pair of real numbers (x, y) written as:

$$z = x + iy, \quad i^2 = -1 \tag{4.1}$$

where $x = \text{Re}(z)$ is the real part, and $y = \text{Im}(z)$ is the imaginary part. The complex conjugate of z is:

$$z^* = \bar{z} = x - iy \tag{4.2}$$

The modulus (absolute magnitude) $|z|$ and argument $\theta = \arg(z)$ are defined geometrically in the **Argand plane**:

$$|z| = \sqrt{zz^*} = \sqrt{x^2 + y^2}, \quad \theta = \arctan\left(\frac{y}{x}\right) \tag{4.3}$$

Theorem 4.1 Euler's Formula and Polar Representation

For any real angle $\theta \in \mathbb{R}$:

$$e^{i\theta} = \cos\theta + i\sin\theta \tag{4.4}$$

Any complex number can thus be written in polar exponential form:

$$z = re^{i\theta} = |z|(\cos\theta + i\sin\theta) \tag{4.5}$$

When $\theta = \pi$, Euler's formula produces the renowned identity connecting fundamental constants:

$$e^{i\pi} + 1 = 0 \tag{4.6}$$

4.1.1 De Moivre's Theorem and Roots of Unity

For any integer $n \in \mathbb{Z}$:

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta) \iff (e^{i\theta})^n = e^{in\theta} \tag{4.7}$$

The n distinct n-th roots of a complex number $z_0 = r_0 e^{i\theta_0}$ are distributed evenly on a circle of radius $\sqrt[n]{r_0}$:

$$z_k = \sqrt[n]{r_0}\exp\left[i\left(\frac{\theta_0 + 2k\pi}{n}\right)\right], \quad k = 0, 1, \ldots, n-1 \tag{4.8}$$

4.2 Physical Applications: Phasors and AC Impedance

In alternating current (AC) electrical networks and wave optics, sinusoidal voltages, currents, and fields of angular frequency ω are represented as the real part of complex rotating vectors (phasors):

$$V(t) = V_0\cos(\omega t + \phi) = \text{Re}\left[\tilde{V}_0 e^{i\omega t}\right], \quad \tilde{V}_0 = V_0 e^{i\phi} \tag{4.9}$$

Since time differentiation corresponds simply to multiplication by $i\omega$:

$$\frac{d}{dt}\left[\tilde{V}_0 e^{i\omega t}\right] = i\omega\left[\tilde{V}_0 e^{i\omega t}\right] \tag{4.10}$$

differential equations governing linear dynamic circuits reduce to algebraic linear equations!

Definition 4.1 Complex Impedance

The generalized complex impedance $\tilde{Z} \equiv \frac{\tilde{V}}{\tilde{I}}$ for fundamental circuit elements is:

- Resistor (R): $\tilde{Z}_R = R$ (current and voltage are in phase).
- Inductor (L): $V = L\frac{dI}{dt} \implies \tilde{Z}_L = i\omega L = \omega L e^{i\pi/2}$ (voltage leads current by $90°$).
- Capacitor (C): $I = C\frac{dV}{dt} \implies \tilde{Z}_C = \frac{1}{i\omega C} = -\frac{i}{\omega C} = \frac{1}{\omega C}e^{-i\pi/2}$ (voltage lags current by $90°$).

Pitfall Alert: Taking Real Parts Before Nonlinear Operations

The complex exponential trick works strictly for **linear** operators (differentiation, integration, linear superposition). When calculating nonlinear physical quantities such as power dissipation $P(t) = V(t)I(t)$ or electromagnetic energy density, one **must take the real parts first** before multiplying:

$$P(t) = \text{Re}[\tilde{V}(t)] \cdot \text{Re}[\tilde{I}(t)] \neq \text{Re}[\tilde{V}(t)\tilde{I}(t)] \tag{4.11}$$

The time-averaged power for sinusoidal steady-state is $\langle P \rangle = \frac{1}{2}\text{Re}[\tilde{V}_0 \tilde{I}_0^*] = V_{\text{rms}} I_{\text{rms}} \cos\phi$.

4.3 Worked Examples and Physical Applications

Example 4.1: Series RLC Circuit Analysis via Complex Impedance

A series RLC circuit is driven by an alternating voltage source $V(t) = V_0 \cos(\omega t)$. Derive the analytical expression for the steady-state current amplitude I_0 and phase angle ϕ between voltage and current.

Picture / Conceptualization: Resistor, inductor, and capacitor are wired in series. The driving voltage phasor is $\tilde{V} = V_0$.

Strategy / Governing Laws: The total equivalent impedance of elements in series is the sum of their individual complex impedances:

$$\tilde{Z}_{\text{total}} = \tilde{Z}_R + \tilde{Z}_L + \tilde{Z}_C = R + i\left(\omega L - \frac{1}{\omega C}\right) \tag{4.12}$$

The complex current phasor is given by Ohm's law in the frequency domain: $\tilde{I} = \frac{\tilde{V}}{\tilde{Z}_{\text{total}}}$.

Calculation / Analytical Solution: 1. Magnitude of total impedance:

$$|\tilde{Z}_{\text{total}}| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} \tag{4.13}$$

2. Phase angle of impedance:

$$\phi = \arg(\tilde{Z}_{\text{total}}) = \arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right) \tag{4.14}$$

3. Current phasor:

$$\tilde{I} = \frac{V_0}{|\tilde{Z}_{\text{total}}|e^{i\phi}} = \frac{V_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} e^{-i\phi} \tag{4.15}$$

4. Physical real current:

$$I(t) = \text{Re}[\tilde{I}e^{i\omega t}] = \frac{V_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \cos(\omega t - \phi) \tag{4.16}$$

Think / Physical Insights: At the resonance frequency $\omega_0 = \frac{1}{\sqrt{LC}}$, the inductive and capacitive reactances cancel exactly ($\omega L = 1/\omega C$). The impedance becomes purely real and minimal ($|\tilde{Z}| = R$), maximizing current amplitude $I_0 = V_0/R$ with $\phi = 0$. ■

Complex Impedance and Resonance Response in Python

```python
import numpy as np
import matplotlib.pyplot as plt

# RLC Circuit Parameters
R = 50.0        # Ohms
L = 100e-3      # 100 mH
C = 10e-6       # 10 uF
V0 = 10.0       # Volts

omega_0 = 1.0 / np.sqrt(L * C)
freq = np.linspace(50, 400, 500)
omega = 2 * np.pi * freq

# Complex impedance calculation
Z = R + 1j * (omega * L - 1.0 / (omega * C))
I_mag = V0 / np.abs(Z)
phase_deg = np.angle(Z, deg=True)

print(f"Resonance Frequency f_0 = {omega_0 / (2*np.pi):.2f} Hz")
print(f"Max Current at resonance = {V0 / R:.3f} A")
```

Example 4.2: Quantum Phase Interference and the Double-Slit Experiment

A quantum particle passes simultaneously through two slits separated by distance d. The probability amplitudes from the two slits at an angle θ are complex numbers $\psi_1 = Ae^{i\phi_1}$ and $\psi_2 = Ae^{i\phi_2}$, where the phase difference is $\delta = \phi_2 - \phi_1 = \frac{2\pi d \sin\theta}{\lambda}$. Calculate (i) the total amplitude $\psi = \psi_1 + \psi_2$, (ii) the probability density $|\psi|^2$, and (iii) the positions of constructive and destructive interference.

Solution:

(i) Total amplitude:

$$\psi = Ae^{i\phi_1}(1 + e^{i\delta}) = Ae^{i\phi_1} \cdot 2\cos\left(\frac{\delta}{2}\right) e^{i\delta/2} = 2A\cos\left(\frac{\delta}{2}\right) e^{i(\phi_1+\delta/2)} \tag{4.17}$$

(ii) Probability density (using modulus squared):

$$\boxed{|\psi|^2 = |2A\cos(\delta/2)|^2 = 4A^2\cos^2\left(\frac{\pi d\sin\theta}{\lambda}\right)} \tag{4.18}$$

(iii) Interference conditions:

$$\text{Constructive: } \cos^2(\delta/2) = 1 \implies \delta = 2m\pi \implies d\sin\theta = m\lambda, \quad m \in \mathbb{Z} \tag{4.19}$$

$$\text{Destructive: } \cos^2(\delta/2) = 0 \implies \delta = (2m+1)\pi \implies d\sin\theta = \left(m + \tfrac{1}{2}\right)\lambda \tag{4.20}$$

Think / Physical Insights: The interference pattern $\propto \cos^2(\delta/2)$ arises entirely from the complex algebra of probability amplitudes. Unlike classical probability (which would give $|\psi_1|^2 + |\psi_2|^2 = 2A^2$, a featureless sum), quantum mechanics predicts an oscillating pattern — an experimentally verified signature of wave-particle duality. The cross term $2\text{Re}[\psi_1\psi_2^*] = 2A^2\cos\delta$ is the quantum interference term. ■

4.4 Summary of Key Formulas

Concept	Formula	Physical Application
Euler's Formula	$e^{i\theta} = \cos\theta + i\sin\theta$	Wave kinematics, rotation operator in quantum mechanics
Complex Impedance	$\tilde{Z} = R + i(\omega L - \frac{1}{\omega C})$	Frequency-domain analysis of AC filters and oscillators
Average Real Power	$\langle P \rangle = \frac{1}{2}\text{Re}[\tilde{V}_0 \tilde{I}_0^*]$	Electrical power transmission, optical energy absorption
Cauchy-Riemann	$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$	2D electrostatics, potential flow around aerofoils
Quantum Interference	$\lvert\psi_1 + \psi_2\rvert^2 = \lvert\psi_1\rvert^2 + \lvert\psi_2\rvert^2 + 2\text{Re}[\psi_1 \psi_2^*]$	Double-slit experiment, thin-film optical coatings

Table 4.1: Summary of complex analysis relations in physical sciences.

4.5 Chapter Exercises

1. **Roots of Unity:** Determine all four roots of the equation $z^4 + 16 = 0$ in polar and Cartesian form, and plot them in the Argand plane.

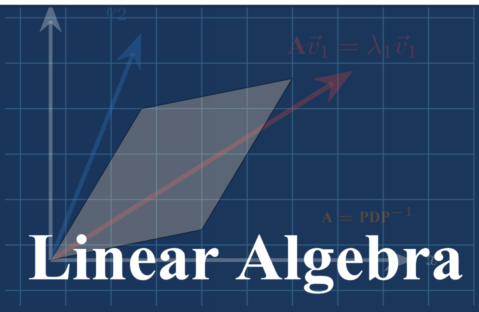
2. **Trigonometric Summation:** Using the geometric series formula with complex exponentials, evaluate the sum $\sum_{k=0}^{n} \cos(k\theta)$.

3. **Parallel RLC Network:** Derive the equivalent complex admittance $\tilde{Y} = 1/\tilde{Z}$ for a parallel combination of R, L, and C, and find the anti-resonance condition.

4. **Cauchy's Integral Formula:** Evaluate the contour integral $\oint_C \frac{e^{iz}}{z^2+1}\, dz$ where C is the circle $|z| = 2$ traversed counterclockwise. Use the residue theorem and state which poles lie inside C.

5. **Phasor Power Factor:** A series RL circuit has resistance $R = 50\ \Omega$ and inductance $L = 0.2$ H. For a source voltage $\tilde{V} = 240e^{i0}$ V at $f = 50$ Hz, calculate the complex power $\tilde{S} = \frac{1}{2}\tilde{V}\tilde{I}^*$ and the power factor $\cos\phi$.

Chapter 5



Linear Algebra & Matrix Mechanics

Chapter 5 Instructional Management Plan

1. Chapter Topics and Core Content

1. Matrix Operations and Special Classes of Matrices
2. The Eigenvalue Problem and Diagonalization
3. Physical Applications: Coupled Oscillators and Normal Modes
4. Worked Examples and Inertia Tensor Transformations
5. Summary of Key Formulas and Chapter Exercises

2. Behavioral Learning Objectives (OBE)

1. Perform matrix algebra, determinant evaluations, and matrix inversions.
2. Solve the secular eigenvalue equation $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$ for eigenvalues and eigenvectors.
3. Diagonalize symmetric and Hermitian operators via similarity transformations.
4. Model coupled mechanical oscillators, normal modes, and inertia tensors.

3. Teaching Methods and Instructional Activities

1. Lecture on linear vector spaces, matrix operators, and spectral decomposition.
2. Workshop deriving normal mode frequencies and eigenvectors for multi-mass systems.
3. Computational lab using Python ‘numpy.linalg‘ for large-scale matrix diagonalization.
4. Case study on principal axes of inertia in rigid body rotational dynamics.

4. Instructional Media and Educational Technology

1. Course textbook: Chapter 5 Linear Algebra & Matrix Mechanics.
2. Lecture slides illustrating 2D/3D geometric transformations and eigenvector shears.
3. Python scripts for matrix operations and coupled oscillator animations.
4. Digital problem sets and matrix mechanics worksheets.

5. Measurement and Learning Assessment

1. In-class engagement and formative assessment of matrix transformation proofs.
2. Evaluation of homework problem sets on eigenvalues and coupled oscillators.
3. Chapter 5 diagnostic examination on matrix diagonalization and normal modes.

5.1 Matrix Operations and Special Classes of Matrices

Linear relationships permeate theoretical physics. Systems of linear equations, spatial rotations, coordinate transformations, and quantum state evolutions are formulated through matrices.

Let $\mathbf{A} \in \mathbb{C}^{n \times n}$ denote an $n \times n$ square matrix with elements A_{ij}.

- **Transpose:** $(\mathbf{A}^T)_{ij} = A_{ji}$.
- **Complex Conjugate Transpose (Hermitian Adjoint):** $\mathbf{A}^\dagger \equiv (\mathbf{A}^*)^T$, so $(A^\dagger)_{ij} = A^*_{ji}$.
- **Trace:** $\mathrm{Tr}(\mathbf{A}) = \sum_{i=1}^{n} A_{ii}$ (invariant under cyclic permutations).
- **Determinant:** Scalar measure of volumetric scaling under the linear transformation $\det(\mathbf{A})$. If $\det(\mathbf{A}) \neq 0$, the matrix is invertible, with inverse satisfying $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$.

Definition 5.1 Classification of Transformation Matrices

A square matrix $\mathbf{A}$ is classified according to its algebraic symmetries:

- **Symmetric:** $\mathbf{A}^T = \mathbf{A}$ (Real entries).
- **Orthogonal:** $\mathbf{A}^T\mathbf{A} = \mathbf{I} \implies \mathbf{A}^{-1} = \mathbf{A}^T$ (Preserves Euclidean vector norms and angles; e.g., 3D rotation matrices).
- **Hermitian:** $\mathbf{A}^\dagger = \mathbf{A}$ (Generalization of symmetric to complex space; all eigenvalues are real, representing quantum observables).
- **Unitary:** $\mathbf{U}^\dagger\mathbf{U} = \mathbf{I} \implies \mathbf{U}^{-1} = \mathbf{U}^\dagger$ (Preserves quantum probability amplitudes; e.g., time evolution operator $U(t) = e^{-iHt/\hbar}$).

5.2 The Eigenvalue Problem and Diagonalization

Definition 5.2 Secular Equation

A non-zero column vector $\vec{v} \neq \vec{0}$ is an **eigenvector** of $\mathbf{A}$ with corresponding **eigenvalue** λ if:

$$\mathbf{A}\vec{v} = \lambda\vec{v} \iff (\mathbf{A} - \lambda\mathbf{I})\vec{v} = \vec{0} \tag{5.1}$$

For non-trivial solutions to exist, the coefficient matrix must be singular, yielding the **characteristic (secular) equation**:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0 \tag{5.2}$$

Expanding this determinant yields an n-th degree polynomial whose roots are the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$.

Theorem 5.1 Spectral Theorem for Hermitian Matrices

If $\mathbf{A}$ is an $n \times n$ Hermitian (or real symmetric) matrix:

1. All n eigenvalues λ_k are strictly real.
2. Eigenvectors corresponding to distinct eigenvalues are mutually orthogonal: $\vec{v}_i^\dagger \vec{v}_j = 0$ for $\lambda_i \neq \lambda_j$.
3. There exists a unitary matrix $\mathbf{U}$ formed by orthonormalized eigenvectors as columns such that:

$$\mathbf{U}^\dagger \mathbf{A} \mathbf{U} = \square = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) \tag{5.3}$$

Pitfall Alert: Non-Commutative Matrix Products

Matrix multiplication is fundamentally non-commutative: in general, $\mathbf{AB} \neq \mathbf{BA}$. The commutator is defined as:

$$[\mathbf{A}, \mathbf{B}] \equiv \mathbf{AB} - \mathbf{BA} \tag{5.4}$$

In quantum mechanics, if two observable operators do not commute ($[\mathbf{A}, \mathbf{B}] \neq \mathbf{0}$), they cannot be measured simultaneously with arbitrary precision (Robertson-Schrödinger uncertainty relation $\Delta A \Delta B \geq \frac{1}{2}|\langle [\mathbf{A}, \mathbf{B}] \rangle|$).

5.3 Physical Applications: Coupled Oscillators and Normal Modes

Consider two identical masses m connected by springs of stiffness k between fixed walls and coupled by a central spring of stiffness k_c. The equations of motion are:

$$m\ddot{x}_1 = -kx_1 - k_c(x_1 - x_2) \tag{5.5}$$
$$m\ddot{x}_2 = -kx_2 - k_c(x_2 - x_1) \tag{5.6}$$

In matrix notation, defining displacement vector $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$:

$$m \begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{pmatrix} + \begin{pmatrix} k + k_c & -k_c \\ -k_c & k + k_c \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{5.7}$$

Assuming harmonic solutions $\vec{x}(t) = \vec{v}e^{i\omega t}$, we obtain the generalized eigenvalue problem $\mathbf{K}\vec{v} = \omega^2 m\vec{v}$.

5.4 Worked Examples and Physical Applications

Example 5.1: Normal Modes of Coupled Masses

Find the normal frequencies ω_1, ω_2 and normal mode vectors for the coupled system with $k_c = k$.

Picture / Conceptualization: Two masses oscillating along a straight line. In mode 1, they oscillate in phase (center spring unstretched). In mode 2, they oscillate in anti-phase (center spring maximally stretched).

Strategy / Governing Laws: The stiffness matrix with $k_c = k$ is $\mathbf{K} = k\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. Solve the characteristic equation $\det(\mathbf{K} - \lambda m\mathbf{I}) = 0$, where $\lambda = \omega^2$.

Calculation / Analytical Solution: 1. Secular determinant:

$$\begin{vmatrix} 2k - m\omega^2 & -k \\ -k & 2k - m\omega^2 \end{vmatrix} = 0 \implies (2k - m\omega^2)^2 - k^2 = 0 \tag{5.8}$$

2. Solving for ω^2:

$$2k - m\omega^2 = \pm k \implies \omega_1^2 = \frac{k}{m}, \quad \omega_2^2 = \frac{3k}{m} \tag{5.9}$$

3. Eigenvectors (Normal Modes): For $\omega_1 = \sqrt{k/m}$:

$$\begin{pmatrix} k & -k \\ -k & k \end{pmatrix}\begin{pmatrix} v_{11} \\ v_{12} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies v_{11} = v_{12} \implies \vec{v}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{5.10}$$

For $\omega_2 = \sqrt{3k/m}$:

$$\begin{pmatrix} -k & -k \\ -k & -k \end{pmatrix}\begin{pmatrix} v_{21} \\ v_{22} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies v_{21} = -v_{22} \implies \vec{v}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} \tag{5.11}$$

Think / Physical Insights: In the symmetric mode $\vec{v}_1$, both masses move in unison ($x_1 = x_2$), so the coupling spring never changes length, yielding the single-mass frequency $\omega_1 = \sqrt{k/m}$. In the antisymmetric mode $\vec{v}_2$, the masses move in opposite directions ($x_1 = -x_2$), subjecting each mass to strong restoring force from the coupling spring and increasing the natural frequency to $\sqrt{3k/m}$. ■

Eigenvalue and Normal Mode Computation in Python

```python
import numpy as np

# System parameters
m = 1.0   # kg
k = 10.0 # N/m

# Stiffness matrix K
K = np.array([[2*k, -k],
```

```
                [-k, 2*k]])

# Eigenvalues and eigenvectors
evals, evecs = np.linalg.eigh(K / m)
frequencies = np.sqrt(evals)

print(f"Normal Mode Frequencies (rad/s): {frequencies}")
print("Orthonormal Eigenvectors (Columns):")
print(evecs)
```

Example 5.2: Diagonalization of a 2D Moment of Inertia Tensor

A thin rigid lamina consists of two equal point masses m located at $\vec{r}_1 = (a, 0, 0)$ and $\vec{r}_2 = (0, b, 0)$. Find the inertia tensor $\mathbf{I}$, determine its eigenvalues (principal moments), and find the principal axes of rotation.

Solution:

The inertia tensor components are $I_{jk} = \sum_\alpha m_\alpha(\delta_{jk}|\vec{r}_\alpha|^2 - r_{\alpha j} r_{\alpha k})$.
For mass 1 at $(a, 0, 0)$: $|\vec{r}_1|^2 = a^2$. For mass 2 at $(0, b, 0)$: $|\vec{r}_2|^2 = b^2$.
Assembling the relevant 2×2 block in the xy-plane:

$$I_{xx} = m(0^2) + m(b^2) = mb^2 \tag{5.12}$$

$$I_{yy} = m(a^2) + m(0^2) = ma^2 \tag{5.13}$$

$$I_{xy} = I_{yx} = -m(a)(0) - m(0)(b) = 0 \tag{5.14}$$

The inertia tensor (in the xy-plane) is diagonal:

$$\mathbf{I} = m \begin{pmatrix} b^2 & 0 \\ 0 & a^2 \end{pmatrix} \tag{5.15}$$

The eigenvalues are $I_1 = mb^2$ (rotation about $\hat{x}$) and $I_2 = ma^2$ (rotation about $\hat{y}$). The eigenvectors are simply $\hat{e}_1 = \hat{x}$ and $\hat{e}_2 = \hat{y}$, confirming that the coordinate axes are already principal axes.

Now consider a perturbed configuration where an additional product of inertia coupling $I_{xy} = -mab/2$ is introduced. The characteristic equation is:

$$\det \begin{pmatrix} mb^2 - \lambda & -mab/2 \\ -mab/2 & ma^2 - \lambda \end{pmatrix} = 0 \implies \lambda^2 - m(a^2 + b^2)\lambda + m^2\left(a^2b^2 - \frac{a^2b^2}{4}\right) = 0 \tag{5.16}$$

$$\boxed{\lambda_{1,2} = \frac{m(a^2 + b^2)}{2} \pm \frac{m}{2}\sqrt{(a^2 - b^2)^2 + a^2b^2}} \tag{5.17}$$

Think / Physical Insights: Diagonalizing $\mathbf{I}$ is physically equivalent to finding a body frame in which the angular momentum $\vec{L} = \mathbf{I}\vec{\omega}$ is always parallel to $\vec{\omega}$. This frame defines the principal axes. For a free rigid body, rotation about the largest and smallest principal moments is stable (Tennis Racket Theorem). ■

5.5 Summary of Key Formulas

Property / Operation	**Formula**	**Physical Meaning**
Characteristic Equation	$\det(\mathbf{A} - \lambda\mathbf{I}) = 0$	Frequencies of normal modes, energy eigenvalues
Hermitian Matrix	$\mathbf{H}^\dagger = \mathbf{H}$	Observables in quantum theory have real spectra
Unitary Transformation	$\mathbf{U}^\dagger\mathbf{U} = \mathbf{I}$	Conservation of total probability and vector lengths
Inertia Tensor Diagonalization	$\mathbf{R}^T\mathbf{I}\mathbf{R} = \mathrm{diag}(I_1, I_2, I_3)$	Finding principal axes of rotation in rigid body mechanics
Spectral Theorem	$\mathbf{A} = \sum_i \lambda_i \lvert\hat{v}_i\rangle\langle\hat{v}_i\rvert$	Any Hermitian operator is a sum of projections onto eigenstates

Table 5.1: Key matrix algebra formulas and their physical interpretations.

5.6 Chapter Exercises

1. **Pauli Spin Matrices:** The Pauli spin matrices in quantum mechanics are $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Prove that each is Hermitian and find their eigenvalues and normalized eigenvectors.

2. **Inertia Tensor:** A system consists of three equal masses m located at $(a, 0, 0)$, $(0, a, a)$, and $(0, 0, a)$. Calculate the inertia tensor components I_{ij} and diagonalize the matrix to determine the principal moments of inertia.

3. **Three-Coupled Oscillators:** Write the 3×3 stiffness matrix for three coupled masses in a ring topology and calculate the normal mode frequencies.

4. **Cayley-Hamilton Theorem:** Every square matrix satisfies its own characteristic equation. Verify this for $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$ by computing $\mathbf{A}^2 - 7\mathbf{A} + 10\mathbf{I}$ and confirming it equals the zero matrix.

5. **Singular Value Decomposition (SVD):** Explain what the singular values σ_i of a real matrix $\mathbf{M}$ represent geometrically. For the matrix $\mathbf{M} = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$, compute $\mathbf{M}^T\mathbf{M}$ and find its eigenvalues, which are σ_i^2.

Chapter 6

Multivariable Differential Calculus

Chapter 6 Instructional Management Plan

1. Chapter Topics and Core Content

1. Partial Derivatives and Clairaut's Symmetry Theorem
2. Total Differentials and Thermodynamic State Functions
3. Directional Derivatives and Tangent Planes
4. Constrained Optimization: Method of Lagrange Multipliers
5. Summary of Key Formulas and Chapter Exercises

2. Behavioral Learning Objectives (OBE)

1. Compute partial derivatives and verify Clairaut's mixed partial symmetry theorem.
2. Formulate total differentials and identify exact differentials in thermodynamics.
3. Determine directional derivatives and tangent planes to 3D scalar surfaces.
4. Solve constrained physical optimization problems using Lagrange multipliers.

3. Teaching Methods and Instructional Activities

1. Theoretical lecture on multivariable surfaces, gradients, and directional slopes.
2. Problem-solving workshop on chain rules and thermodynamic Maxwell relations.
3. Computational lab generating 3D surface and contour plots with Python.
4. Collaborative analysis of constrained energy minimization using Lagrange multipliers.

4. Instructional Media and Educational Technology

1. Course textbook: Chapter 6 Multivariable Differential Calculus.
2. Interactive 3D visual slides of saddle points, local extrema, and tangent planes.
3. Python scripts for gradient vector field visualizations and contour plots.
4. Homework worksheets on thermodynamic differentials and optimization.

5. Measurement and Learning Assessment

1. Formative evaluation of in-class problem solving and concept checks.
2. Homework grading on directional derivatives and Lagrange multiplier constraints.
3. Chapter 6 diagnostic quiz on multivariable differentiation and physical extrema.

6.1 Partial Derivatives and Clairaut's Symmetry Theorem

Physical fields in classical mechanics, thermodynamics, and electromagnetism depend on multiple spatial and temporal coordinates (x, y, z, t). For a scalar function $f(x, y)$, the **partial derivative** with respect to x measures the rate of change while holding y strictly constant:

$$\frac{\partial f}{\partial x} \equiv \lim_{\Delta x \to 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x} \tag{6.1}$$

Theorem 6.1 Clairaut's Theorem on Mixed Partials

If the partial derivatives $\frac{\partial^2 f}{\partial x \partial y}$ and $\frac{\partial^2 f}{\partial y \partial x}$ exist and are continuous on an open disk containing point (x_0, y_0), then the order of differentiation is commutative:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} \tag{6.2}$$

In thermodynamics, this mathematical symmetry generates the powerful **Maxwell relations** connecting pressure, volume, temperature, and entropy.

6.2 Total Differentials and Thermodynamic State Functions

The **total differential** of a function $f(x, y, z)$ represents the first-order change induced by arbitrary infinitesimal increments (dx, dy, dz):

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz = \nabla f \cdot d\vec{r} \tag{6.3}$$

Definition 6.1 Exact vs. Inexact Differentials

A differential form $\delta Q = P(x, y)dx + Q(x, y)dy$ is **exact** if it can be written as the total differential of a state function $f(x, y)$, which requires:

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \tag{6.4}$$

In thermodynamics, internal energy $dU = TdS - PdV$ is an exact differential (a state function independent of path), whereas heat δQ and work δW are inexact differentials whose integrals depend explicitly on the thermodynamic process path.

Pitfall Alert: Thermodynamic Notation and Constrained Variables

In thermodynamics, writing $\frac{\partial U}{\partial T}$ is ambiguous because internal energy can be expressed in terms of (T, V) or (T, P). One must always specify the held-constant variable explicitly using subscripts:

$$\left(\frac{\partial U}{\partial T}\right)_V \neq \left(\frac{\partial U}{\partial T}\right)_P \tag{6.5}$$

6.3 Directional Derivatives and Tangent Planes

The directional derivative of $f(x, y, z)$ along a unit direction vector $\hat{u} = u_x\hat{i} + u_y\hat{j} + u_z\hat{k}$ is:

$$D_{\hat{u}}f = \nabla f \cdot \hat{u} = |\nabla f| \cos\theta \tag{6.6}$$

The rate of change is maximized when $\hat{u}$ points along the gradient ($\theta = 0$), where $D_{\hat{u}}f = |\nabla f|$. It vanishes when $\hat{u}$ is tangent to the level surface ($\theta = 90°$).

For a level surface defined by $F(x, y, z) = 0$, the gradient vector $\vec{n} = \nabla F(x_0, y_0, z_0)$ is strictly normal to the tangent plane at (x_0, y_0, z_0). The equation of the **tangent plane** is:

$$\nabla F(x_0, y_0, z_0) \cdot (\vec{r} - \vec{r}_0) = 0 \iff F_x(x - x_0) + F_y(y - y_0) + F_z(z - z_0) = 0 \tag{6.7}$$

6.4 Constrained Optimization: The Method of Lagrange Multipliers

Physical systems frequently settle into equilibrium states that minimize potential energy or maximize entropy subject to geometric or conservation constraints.

Theorem 6.2 Lagrange Multiplier Principle

To find the extrema of an objective function $f(x, y, z)$ subject to an equality constraint $g(x, y, z) = 0$, the gradients of f and g must be parallel at the extremum point:

$$\nabla f = \lambda \nabla g \tag{6.8}$$

where λ is the **Lagrange multiplier**. Along with the constraint $g(x, y, z) = 0$, this yields four equations in four unknowns (x, y, z, λ).

6.5 Worked Examples and Physical Applications

Example 6.1: Minimizing Electrostatic Potential Energy on a Constraint Surface

A charged particle moves on the elliptical cylinder constraint surface $g(x, y) = x^2 + 4y^2 - 4 = 0$ in the presence of an electrostatic potential $V(x, y) = x^2 + y^2 - 2x$. Determine the location of the constrained equilibrium point that minimizes potential energy.

Picture / Conceptualization: The particle is physically confined to an elliptical wire loop. The stable equilibrium position corresponds to the lowest point of the potential energy landscape on this ellipse.

Strategy / Governing Laws: Apply the Lagrange multiplier condition $\nabla V = \lambda \nabla g$ with the ellipse constraint $g(x, y) = 0$.

Calculation / Analytical Solution: 1. Compute gradients:

$$\nabla V = (2x - 2)\hat{i} + 2y\hat{j} \tag{6.9}$$

$$\nabla g = 2x\hat{i} + 8y\hat{j} \tag{6.10}$$

2. Set up component equations:

$$2x - 2 = 2\lambda x \implies x(1-\lambda) = 1 \implies x = \frac{1}{1-\lambda} \tag{6.11}$$

$$2y = 8\lambda y \implies y(1-4\lambda) = 0 \tag{6.12}$$

3. Case Analysis:

- **Case A:** $y = 0$. Substituting into constraint $x^2 + 4(0) = 4 \implies x = \pm 2$.
  - At $(2,0)$: $V(2,0) = 4 + 0 - 4 = 0$.
  - At $(-2,0)$: $V(-2,0) = 4 + 0 + 4 = 8$.
- **Case B:** $1 - 4\lambda = 0 \implies \lambda = 1/4$. Then $x = \frac{1}{1-1/4} = \frac{4}{3}$. From constraint: $(4/3)^2 + 4y^2 = 4 \implies \frac{16}{9} + 4y^2 = 4 \implies 4y^2 = \frac{20}{9} \implies y = \pm\frac{\sqrt{5}}{3}$. Evaluating potential:

$$V\left(\frac{4}{3}, \pm\frac{\sqrt{5}}{3}\right) = \left(\frac{4}{3}\right)^2 + \frac{5}{9} - 2\left(\frac{4}{3}\right) = \frac{16+5-24}{9} = -\frac{3}{9} = -\frac{1}{3} \tag{6.13}$$

Think / Physical Insights: Comparing the values: $V(4/3, \pm\sqrt{5}/3) = -1/3$ is strictly lower than $V(2,0) = 0$ and $V(-2,0) = 8$. Thus, there are two symmetric stable equilibrium points at $(4/3, \pm\sqrt{5}/3)$. ■

Lagrange Multiplier Contour Plot in Python

```python
import numpy as np

# Verify potential values on the ellipse x = 2*cos(t), y = sin(t)
t = np.linspace(0, 2*np.pi, 1000)
x = 2 * np.cos(t)
y = np.sin(t)
V = x**2 + y**2 - 2*x

min_idx = np.argmin(V)
print(f"Numerical Minimum V = {V[min_idx]:.4f} at x = {x[min_idx]:.4f},
    y = {y[min_idx]:.4f}")
print(f"Analytical Minimum: -1/3 = {-1/3:.4f}, x = 4/3 = {4/3:.4f}, y =
    sqrt(5)/3 = {np.sqrt(5)/3:.4f}")
```

Example 6.2: Fourier's Heat Law and the Temperature Gradient

A cylindrical aluminium rod of length L has its left end maintained at $T_0 = 100$ °C and its right end at $T_L = 20$ °C. At steady state, the temperature profile along the rod is $T(x) = T_0 + (T_L - T_0)\frac{x}{L}$. Calculate (i) the temperature gradient ∇T, (ii) the heat flux vector $\vec{q} = -k\nabla T$ with thermal conductivity $k = 237\ \mathrm{W\,m^{-1}K^{-1}}$, and (iii) verify that $\nabla^2 T = 0$ at steady state.

Solution:

(i) Temperature gradient in the x-direction:

$$\nabla T = \frac{dT}{dx}\hat{\imath} = \frac{T_L - T_0}{L}\hat{\imath} = \frac{20-100}{L}\hat{\imath} = -\frac{80}{L}\hat{\imath} \quad [\mathrm{K\,m^{-1}}] \tag{6.14}$$

(ii) Heat flux vector (Fourier's law):

$$\vec{q} = -k\nabla T = -237 \times \left(-\frac{80}{L}\right)\hat{\imath} = \frac{18960}{L}\hat{\imath} \quad [\mathrm{W\,m^{-2}}] \tag{6.15}$$

The positive sign confirms heat flows in the $+x$ direction (from hot to cold).
(iii) Laplacian verification:

$$\nabla^2 T = \frac{d^2T}{dx^2} = \frac{d}{dx}\left(-\frac{80}{L}\right) = 0 \quad \checkmark \tag{6.16}$$

This confirms the linear profile satisfies Laplace's equation $\nabla^2 T = 0$, as expected for steady-state heat conduction with no internal heat sources.
Think / Physical Insights: At steady state, the heat equation $\frac{\partial T}{\partial t} = \alpha\nabla^2 T$ reduces to Laplace's equation $\nabla^2 T = 0$. The temperature profile is harmonic — it has no local maxima or minima in the interior (maximum principle). This same mathematical structure governs electrostatic potential, gravitational potential, and steady-state diffusion. ■

6.6 Summary of Key Formulas

Theorem / Concept	Formula	Physical Application
Clairaut's Symmetry	$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$	Maxwell relations in thermodynamics, integrability of potentials
Total Differential	$df = \sum_i \frac{\partial f}{\partial x_i} dx_i$	First law of thermodynamics, propagation of experimental errors
Directional Derivative	$D_{\hat{u}} f = \nabla f \cdot \hat{u}$	Temperature gradients in heat conduction, pressure gradients in meteorology
Lagrange Multipliers	$\nabla f = \sum_k \lambda_k \nabla g_k$	Microcanonical entropy maximization, constraint forces in D'Alembert mechanics
Fourier's Heat Law	$\vec{q} = -k\nabla T$	Heat conduction, Fick's diffusion, Darcy's porous flow

Table 6.1: Key relations in multivariable differential calculus and thermodynamics.

6.7 Chapter Exercises

1. **Thermodynamic Maxwell Relation:** From the Helmholtz free energy $F = U - TS$ with $dF = -SdT - PdV$, prove that $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$.

2. **Tangent Plane to an Ellipsoid:** Find the equation of the tangent plane to the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ at an arbitrary point (x_0, y_0, z_0).

3. **Maximum Area with Fixed Perimeter:** Use Lagrange multipliers to prove that among all rectangles of fixed perimeter P, the square encloses the maximum area.

4. **Gradient in Spherical Coordinates:** A spherically symmetric gravitational potential is $\Phi(r) = -GM/r$. Calculate $\vec{g} = -\nabla\Phi$ in spherical coordinates and verify that $\nabla \cdot \vec{g} = 0$ for $r \neq 0$ (Gauss's law in vacuum).

5. **Second Derivative Test in 2D:** The function $f(x, y) = x^3 + y^3 - 3xy$ has a critical point structure. Find all critical points, compute the Hessian determinant $D = f_{xx}f_{yy} - f_{xy}^2$ at each, and classify each as a local minimum, maximum, or saddle point.

Chapter 7

Multiple Integrals & Field Theorems

Chapter 7 Instructional Management Plan

1. Chapter Topics and Core Content

1. Multiple Integrals and the Jacobian Transformation
2. Line and Surface Integrals in Physics
3. The Fundamental Theorems of Vector Integral Calculus
4. Worked Examples: Gauss's Divergence and Stokes' Theorems
5. Summary of Key Formulas and Chapter Exercises

2. Behavioral Learning Objectives (OBE)

1. Transform variables in double and triple integrals using the Jacobian determinant.
2. Compute line integrals for work and circulation along spatial trajectories.
3. Apply Gauss's Divergence theorem to relate volume divergence to surface flux.
4. Apply Stokes' theorem to relate surface vorticity to closed-path circulation.

3. Teaching Methods and Instructional Activities

1. Lecture on line, surface, and volume integrals and integral theorem unifications.
2. Workshop calculating Jacobian scaling factors across curvilinear coordinates.
3. Computational lab computing numerical multiple integrals with SciPy integrate.
4. Group seminar connecting vector integral theorems to Maxwell's field equations.

4. Instructional Media and Educational Technology

1. Course textbook: Chapter 7 Multiple Integrals & Field Theorems.
2. 3D visual animations of flux penetrating closed Gaussian surfaces.
3. SciPy Python integration scripts for multi-dimensional volume integrals.
4. Analytical problem sets and integral vector calculus study guides.

5. Measurement and Learning Assessment

1. In-class problem solving and conceptual checks on boundary orientations.
2. Grading of homework problem sets on flux integrals and Stokes' circulation.
3. Chapter 7 diagnostic quiz covering integral transformation theorems.

7.1 Multiple Integrals and the Jacobian Transformation

Physical quantities such as total mass $M = \iiint \rho(\vec{r})\,dV$, center of mass $\vec{r}_{\text{cm}} = \frac{1}{M}\iiint \vec{r}\rho\,dV$, and moments of inertia require integration over two- and three-dimensional spatial manifolds.

When changing integration variables from (x, y, z) to curvilinear coordinates (u, v, w), the volume transformation is scaled by the absolute value of the **Jacobian determinant**:

$$J \equiv \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} \tag{7.1}$$

The integral transformation law is:

$$\iiint_V f(x, y, z)\,dx\,dy\,dz = \iiint_{V'} f(x(u, v, w), y(u, v, w), z(u, v, w))\,|J|\,du\,dv\,dw \tag{7.2}$$

For orthogonal curvilinear coordinates, the Jacobian equals the product of the scale factors: $|J| = h_1h_2h_3$. Specifically:

- Cylindrical: $|J| = \rho \implies dV = \rho\,d\rho\,d\phi\,dz$.
- Spherical: $|J| = r^2 \sin\theta \implies dV = r^2 \sin\theta\,dr\,d\theta\,d\phi$.

7.2 The Fundamental Theorems of Vector Integral Calculus

Theorem 7.1 Green's Theorem in the Plane

Let C be a positively oriented (counterclockwise), piecewise-smooth, simple closed curve enclosing region $D \subset \mathbb{R}^2$. If $P(x, y)$ and $Q(x, y)$ have continuous partial derivatives on D:

$$\oint_C (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA \tag{7.3}$$

Theorem 7.2 Stokes' Theorem (Circulation Theorem)

Let S be an oriented piecewise-smooth surface bounded by a closed contour $C = \partial S$ oriented by the right-hand rule with respect to the unit normal $\hat{n}$. For any continuously differentiable vector field $\vec{A}$:

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\nabla \times \vec{A}) \cdot d\vec{a} \tag{7.4}$$

In electrodynamics, applying Stokes' theorem to Faraday's law $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ generates the macroscopic induction law:

$$\mathcal{E} = \oint_C \vec{E} \cdot d\vec{r} = -\frac{d}{dt} \iint_S \vec{B} \cdot d\vec{a} = -\frac{d\Phi_B}{dt} \tag{7.5}$$

Theorem 7.3 Gauss's Divergence Theorem

Let V be a compact three-dimensional region bounded by a closed surface $S = \partial V$ oriented with an outward-pointing unit normal $\hat{n}$. For any continuously differentiable vector field $\vec{A}$:

$$\oiint_S \vec{A} \cdot d\vec{a} = \iiint_V (\nabla \cdot \vec{A})\, dV \tag{7.6}$$

Gauss's divergence theorem bridges the microscopic differential form of Gauss's law $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$ and its macroscopic integral form $\oiint_S \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$.

Pitfall Alert: Orientation and Right-Hand Rule Conventions

In Stokes' theorem, the boundary curve C must be traversed such that when walking along C with one's head pointing in the direction of the surface normal $\hat{n}$, the surface S lies to one's left. In Gauss's theorem, S must be a **closed** surface, and $d\vec{a} = \hat{n}\, da$ must point **outward**.

7.3 Worked Examples and Physical Applications

Example 7.1: Verification of Gauss's Divergence Theorem

Let $\vec{F}(x,y,z) = x\hat{i} + y\hat{j} + z\hat{k} = \vec{r}$. Verify Gauss's divergence theorem over a solid sphere of radius R centered at the origin.
Picture / Conceptualization: The vector field points radially outward everywhere from the origin with magnitude proportional to distance. The bounding surface is the sphere S of radius R.
Strategy / Governing Laws: Evaluate both sides of $\oiint_S \vec{F} \cdot d\vec{a} = \iiint_V (\nabla \cdot \vec{F})\, dV$ independently and compare the results.
Calculation / Analytical Solution: 1. Volume integral: Compute the divergence of the position vector:

$$\nabla \cdot \vec{r} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3 \tag{7.7}$$

Integrate over the spherical volume:

$$\iiint_V (\nabla \cdot \vec{F})\, dV = 3 \iiint_V dV = 3\left(\frac{4}{3}\pi R^3\right) = 4\pi R^3 \tag{7.8}$$

2. Surface integral: On the spherical surface of radius R, the outward unit normal is $\hat{n} = \hat{r}$, and the vector field is $\vec{F} = R\hat{r}$. The surface area element is $d\vec{a} = \hat{r}\, da$.

$$\oiint_S \vec{F} \cdot d\vec{a} = \oiint_S (R\hat{r}) \cdot (\hat{r}\, da) = R \oiint_S da = R(4\pi R^2) = 4\pi R^3 \tag{7.9}$$

Think / Physical Insights: Both calculations yield precisely $4\pi R^3$. The divergence theorem converted a two-dimensional spherical surface integral into a trivial volume integral of a constant scalar divergence! ■

Numerical Verification of Flux through a Sphere

```python
import numpy as np

# Monte Carlo integration over unit sphere
N = 1000000
pts = np.random.uniform(-1, 1, (N, 3))
r_sq = np.sum(pts**2, axis=1)
inside = pts[r_sq <= 1.0]

vol_sphere = (len(inside) / N) * 8.0 # Box volume is 2^3 = 8
div_F = 3.0
vol_integral = div_F * vol_sphere

exact = 4.0 * np.pi * (1.0)**3
print(f"Numerical Volume Integral: {vol_integral:.4f}")
print(f"Exact Analytical Flux:     {exact:.4f}")
```

Example 7.2: Gauss's Law for a Uniformly Charged Sphere

A solid sphere of radius R carries a uniform charge density ρ_0 [$\mathrm{C\,m^{-3}}$]. Using the divergence theorem, (i) derive the electric field $\vec{E}$ for $r < R$ (interior) and $r > R$ (exterior), and (ii) verify that Gauss's law $\oiint_S \vec{E} \cdot d\vec{a} = Q_{\text{enc}}/\epsilon_0$ holds at both a spherical Gaussian surface of radius $r < R$ and $r = 2R$.

Solution:

By spherical symmetry, $\vec{E} = E(r)\hat{r}$. Choose a concentric Gaussian sphere of radius r.

(i) Interior region ($r < R$): The enclosed charge is $Q_{\text{enc}} = \rho_0 \cdot \frac{4}{3}\pi r^3$.

$$\oiint_S \vec{E}\cdot d\vec{a} = E(r)\cdot 4\pi r^2 = \frac{Q_{\text{enc}}}{\epsilon_0} = \frac{\rho_0}{\epsilon_0}\cdot\frac{4\pi r^3}{3} \implies \boxed{E(r) = \frac{\rho_0 r}{3\epsilon_0}, \quad r \le R} \tag{7.10}$$

(ii) Exterior region ($r > R$): The total charge is $Q = \rho_0 \cdot \frac{4}{3}\pi R^3$.

$$E(r)\cdot 4\pi r^2 = \frac{Q}{\epsilon_0} = \frac{\rho_0 R^3}{3\epsilon_0}\cdot\frac{4\pi}{1} \implies \boxed{E(r) = \frac{\rho_0 R^3}{3\epsilon_0 r^2} = \frac{Q}{4\pi\epsilon_0 r^2}, \quad r > R} \tag{7.11}$$

Verification at $r = 2R$:

$$\Phi_E = E(2R)\cdot 4\pi(2R)^2 = \frac{\rho_0 R^3}{3\epsilon_0 (2R)^2}\cdot 16\pi R^2 = \frac{4\pi\rho_0 R^3}{3\epsilon_0} = \frac{Q}{\epsilon_0} \quad \checkmark \tag{7.12}$$

Think / Physical Insights: The interior field grows linearly with r (analogous to a restoring Hookean force), while the exterior field follows an inverse-square law. This is Coulomb's law in disguise. The same mathematical structure appears in Newtonian gravity inside a massive shell (Newton's shell theorem), confirming that the divergence theorem is the unifying mathematical framework behind both laws. ■

7.4 Summary of Key Formulas

Theorem	Integral Equation	Physical Meaning
Green's Theorem	$\oint_{\partial D}(Pdx + Qdy) = \iint_D(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y})dA$	Planar circulation equals enclosed vorticity flux
Stokes' Theorem	$\oint_{\partial S} \vec{A}\cdot d\vec{r} = \iint_S(\nabla\times\vec{A})\cdot d\vec{a}$	Faraday's induction and Ampère's circuital laws
Divergence Theorem	$\oiint_{\partial V} \vec{A}\cdot d\vec{a} = \iiint_V(\nabla\cdot\vec{A})dV$	Gauss's law for electrostatic and gravitational flux
Jacobian Rule	$dV = \lvert\det J\rvert\, du_1 du_2 du_3$	Geometric volume distortion under curvilinear transforms
Gauss's Law	$\oiint_S \vec{E}\cdot d\vec{a} = Q_{\text{enc}}/\epsilon_0$	Electric flux through closed surface = enclosed charge/ϵ_0

Table 7.1: Summary of fundamental integral transformation theorems.

7.5 Chapter Exercises

1. **Stokes' Theorem on a Paraboloid:** Verify Stokes' theorem for $\vec{A} = -y\hat{\imath} + x\hat{\jmath} + z\hat{k}$ over the paraboloid surface $z = 1 - x^2 - y^2$ ($z \geq 0$) bounded by the unit circle in the xy-plane.

2. **Gravitational Field Divergence:** Show that for the Newtonian gravitational field $\vec{g} = -\frac{GM}{r^2}\hat{r}$, $\oiint_S \vec{g} \cdot d\vec{a} = -4\pi GM$ for any closed surface enclosing the origin.

3. **Jacobian in Elliptic Coordinates:** Calculate the Jacobian determinant for planar elliptic coordinates $x = a \cosh u \cos v$, $y = a \sinh u \sin v$.

4. **Ampère's Law via Stokes:** An infinite straight wire carries steady current I. Using Stokes' theorem with $\vec{B} = \frac{\mu_0 I}{2\pi\rho}\hat{\phi}$ in cylindrical coordinates, verify Ampère's law $\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I$ for a circular loop of radius ρ around the wire.

5. **Solid Angle via Divergence:** Use the identity $\nabla \cdot (\vec{r}/r^3) = 4\pi\delta^3(\vec{r})$ and the divergence theorem to prove that the solid angle subtended by a closed surface S at a point P inside it equals 4π.

Chapter 8

First-Order Ordinary Differential Equations

Chapter 8 Instructional Management Plan

1. Chapter Topics and Core Content

1. Classification and General Structure of ODEs
2. Fundamental Analytical Solution Methods
3. Physical Applications: Atmospheric Drag and Terminal Velocity
4. Worked Examples: RC Circuits and Radioactive Decay
5. Summary of Key Formulas and Chapter Exercises

2. Behavioral Learning Objectives (OBE)

1. Classify differential equations by order, linearity, and initial conditions.
2. Solve separable, linear with integrating factors, and exact differential equations.
3. Model mechanical free-fall under drag and transient charging in RC circuits.
4. Solve initial-value problems numerically using SciPy `solve_ivp`.

3. Teaching Methods and Instructional Activities

1. Lecture on slope fields, integrating factors, and existence-uniqueness theorems.
2. Workshop deriving transient responses of electric circuits and Newton's law of cooling.
3. Computational laboratory implementing Euler and Runge-Kutta numerical solvers in Python.
4. Case study on velocity limits and aerodynamic drag in skydiving physics.

4. Instructional Media and Educational Technology

1. Course textbook: Chapter 8 First-Order Ordinary Differential Equations.
2. Visual slope field plots and dynamic direction field generators.
3. Python scripts using SciPy `odeint` and `solve_ivp`.
4. Homework worksheets and analytical differential equation drills.

5. Measurement and Learning Assessment

1. Formative in-class assessment on integrating factor calculations.
2. Weekly problem set grading on physical modeling and terminal velocity ODEs.
3. Chapter 8 diagnostic quiz testing analytical solutions and numerical modeling.

8.1 Classification and General Structure of ODEs

In dynamical systems, Newton's second law $\vec{F} = m\frac{d^2\vec{r}}{dt^2}$, radioactive decay $\frac{dN}{dt} = -\lambda N$, and circuit laws are formulated as differential equations relating an unknown function to its derivatives.

An ordinary differential equation (ODE) contains only one independent variable. The **order** is the highest derivative present. A first-order ODE can be written in explicit form:

$$\frac{dy}{dx} = f(x, y) \quad \text{or differential form} \quad M(x, y)\,dx + N(x, y)\,dy = 0 \tag{8.1}$$

8.2 Fundamental Analytical Solution Methods

8.2.1 Separable Differential Equations

If the right-hand side factors into functions of x and y separately:

$$\frac{dy}{dx} = g(x)h(y) \implies \frac{1}{h(y)}\,dy = g(x)\,dx \tag{8.2}$$

Integrating both sides yields the general solution:

$$\int \frac{1}{h(y)}\,dy = \int g(x)\,dx + C \tag{8.3}$$

8.2.2 First-Order Linear ODEs and Integrating Factors

A first-order equation is **linear** if it can be arranged in the canonical standard form:

$$\frac{dy}{dx} + P(x)y = Q(x) \tag{8.4}$$

Theorem 8.1 The Integrating Factor Method

Multiplying both sides by the **integrating factor** $\mu(x) \equiv \exp\left(\int P(x)\,dx\right)$ transforms the left-hand side into the derivative of a product:

$$\frac{d}{dx}\left[\mu(x)y\right] = \mu(x)Q(x) \tag{8.5}$$

Integrating directly yields the general solution:

$$y(x) = \frac{1}{\mu(x)}\left[\int \mu(x)Q(x)\,dx + C\right] \tag{8.6}$$

Pitfall Alert: Ensuring Standard Normal Form

Before computing the integrating factor $\mu(x) = e^{\int P(x)dx}$, the differential equation **must** have a leading coefficient of unity for $\frac{dy}{dx}$. If given $x\frac{dy}{dx} + 2y = 4x^2$, one must first divide by x to identify $P(x) = 2/x$, giving $\mu(x) = e^{\int (2/x)dx} = x^2$.

8.3 Physical Applications: Atmospheric Drag and Terminal Velocity

Consider a body of mass m falling vertically under gravity g through a fluid. At high Reynolds numbers (e.g., a skydiver in air), the drag force is proportional to the square of velocity:

$$F_{\text{drag}} = -cv^2, \quad c = \frac{1}{2}\rho_{\text{air}} C_D A \tag{8.7}$$

Newton's second law gives:

$$m\frac{dv}{dt} = mg - cv^2 \implies \frac{dv}{dt} = g\left(1 - \frac{v^2}{v_t^2}\right) \tag{8.8}$$

where $v_t \equiv \sqrt{\frac{mg}{c}}$ is the **terminal velocity**, reached when drag balances gravity ($dv/dt = 0$).

8.4 Worked Examples and Physical Applications

Example 8.1: Analytical Solution of Skydiver with Quadratic Drag

Assuming the skydiver drops from rest ($v(0) = 0$), solve the nonlinear ODE for $v(t)$ and compute the time required to reach 95% of terminal velocity.

Picture / Conceptualization: The skydiver accelerates downward. As speed increases, opposing air drag grows quadratically until it equals gravitational weight.

Strategy / Governing Laws: The equation is separable: $\frac{dv}{1-(v/v_t)^2} = g\,dt$. Integrate using partial fractions or the hyperbolic substitution $\int \frac{du}{1-u^2} = \tanh^{-1}(u)$.

Calculation / Analytical Solution: 1. Separate variables:

$$\int \frac{dv}{v_t^2 - v^2} = \frac{g}{v_t^2}\int dt \tag{8.9}$$

Using the standard integral $\int \frac{dv}{v_t^2-v^2} = \frac{1}{2v_t}\ln\left|\frac{v_t+v}{v_t-v}\right|$:

$$\frac{1}{2v_t}\ln\left(\frac{v_t+v}{v_t-v}\right) = \frac{gt}{v_t^2} + C \tag{8.10}$$

2. Initial condition $v(0) = 0 \implies \ln(1) = 0 \implies C = 0$. 3. Exponentiating both sides:

$$\frac{v_t+v}{v_t-v} = e^{2gt/v_t} \implies v(t) = v_t\left(\frac{e^{2gt/v_t}-1}{e^{2gt/v_t}+1}\right) = v_t\tanh\left(\frac{gt}{v_t}\right) \tag{8.11}$$

4. Time to reach $v = 0.95v_t$:

$$\tanh\left(\frac{gt_{95}}{v_t}\right) = 0.95 \implies \frac{gt_{95}}{v_t} = \tanh^{-1}(0.95) = \frac{1}{2}\ln\left(\frac{1.95}{0.05}\right) = \frac{1}{2}\ln(39) \approx 1.832 \tag{8.12}$$

Therefore:

$$t_{95} \approx 1.832\frac{v_t}{g} \tag{8.13}$$

For a typical human skydiver with $v_t \approx 54$ m/s (195 km/h) and $g = 9.8$ m/s^2:

$$t_{95} \approx 1.832 \left(\frac{54}{9.8}\right) \approx 10.1 \text{ seconds} \tag{8.14}$$

Think / Physical Insights: Because $\tanh(x) \to 1$ as $x \to \infty$, the terminal velocity is approached asymptotically. In approximately 10 seconds, the skydiver reaches over 95% of their final steady speed. ■

Nonlinear ODE Integration with SciPy solve_ivp

```python
import numpy as np
from scipy.integrate import solve_ivp
import matplotlib.pyplot as plt

g = 9.8
vt = 54.0

def ode_drag(t, v):
    return g * (1.0 - (v / vt)**2)

sol = solve_ivp(ode_drag, [0, 20], [0.0], t_eval=np.linspace(0, 20,
    200))

v_analytical = vt * np.tanh(g * sol.t / vt)
max_error = np.max(np.abs(sol.y[0] - v_analytical))

print(f"Terminal velocity: {vt:.1f} m/s")
print(f"Maximum discrepancy between numerical and analytical solutions:
     {max_error:.2e} m/s")
```

Example 8.2: Logistic Population Growth — Nonlinear First-Order ODE

A bacterial population $N(t)$ in a nutrient-limited environment obeys the logistic equation:

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right) \tag{8.15}$$

where $r = 0.5$ h^{-1} is the intrinsic growth rate and $K = 10^6$ cells is the carrying capacity. Given $N(0) = 1000$ cells, (i) solve the ODE analytically, (ii) find the time $t_{1/2}$ when $N = K/2$, and (iii) determine $\lim_{t\to\infty} N(t)$.

Solution:

(i) Separation of variables:

$$\frac{dN}{N(1 - N/K)} = r\,dt \implies \int \left(\frac{1}{N} + \frac{1}{K - N}\right) dN = r\,dt \tag{8.16}$$

Integrating both sides:

$$\ln N - \ln(K - N) = rt + C \implies \frac{N}{K - N} = Ae^{rt}, \quad A = \frac{N_0}{K - N_0} \tag{8.17}$$

Solving for $N(t)$:

$$\boxed{N(t) = \frac{K}{1 + \left(\dfrac{K - N_0}{N_0}\right) e^{-rt}} = \frac{10^6}{1 + 999\, e^{-0.5t}}} \tag{8.18}$$

(ii) Time to reach $N = K/2$:

$$\frac{K}{2} = \frac{K}{1 + Ce^{-rt_{1/2}}} \implies Ce^{-rt_{1/2}} = 1 \implies t_{1/2} = \frac{\ln(K/N_0 - 1)}{r} = \frac{\ln 999}{0.5} \approx 13.8 \text{ h} \tag{8.19}$$

(iii) Long-time limit: $\lim_{t\to\infty} N(t) = K = 10^6$ cells (carrying capacity is a stable equilibrium).

Think / Physical Insights: The inflection point (maximum growth rate) occurs at $N = K/2$. The logistic model appears in epidemiology (SIR model), tumour growth kinetics, the adoption curve of technologies (Bass diffusion model), and enzyme kinetics (Michaelis-Menten equation). The Bernoulli substitution $u = N^{1-1} = N^0$ is not needed here — separation of variables suffices due to the product structure. ■

Logistic Growth Numerical vs. Analytical Comparison

```python
import numpy as np
from scipy.integrate import solve_ivp

r, K, N0 = 0.5, 1e6, 1000.0

def logistic(t, N):
    return r * N * (1 - N / K)

sol = solve_ivp(logistic, [0, 40], [N0], dense_output=True)
t_test = np.linspace(0, 40, 5)
N_analytical = K / (1 + ((K - N0)/N0) * np.exp(-r * t_test))
N_numerical  = sol.sol(t_test)[0]

for t, Na, Nn in zip(t_test, N_analytical, N_numerical):
    print(f"t={t:4.0f}h  Analytical={Na:.0f}  Numerical={Nn:.0f}  Err={
    abs(Na-Nn):.2e}")
```

8.5 Summary of Key Formulas

Equation Type	Standard Form	Solution Method
Separable	$\frac{dy}{dx} = g(x)h(y)$	Direct integration: $\int \frac{dy}{h(y)} = \int g(x)dx + C$
Linear 1st-Order	$\frac{dy}{dx} + P(x)y = Q(x)$	Integrating factor: $\mu(x) = \exp\left(\int P(x)dx\right)$
Exact Equation	$M\,dx + N\,dy = 0$, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$	Potential function: $\phi(x, y) = C$
RC Circuit Discharge	$R\frac{dQ}{dt} + \frac{Q}{C} = 0$	Exponential relaxation: $Q(t) = Q_0 e^{-t/\tau}$, $\tau = RC$
Logistic Equation	$\frac{dN}{dt} = rN(1 - N/K)$	Separation: $N(t) = K/[1+(K/N_0 - 1)e^{-rt}]$

Table 8.1: Classification and solution methods for first-order ordinary differential equations.

8.6 Chapter Exercises

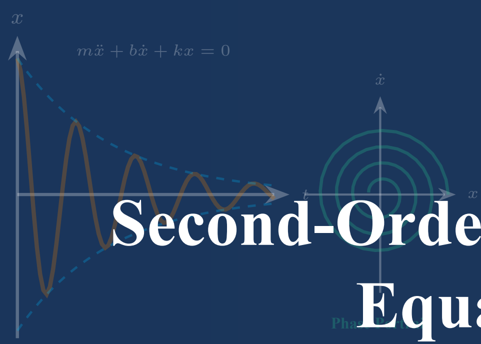
1. **Series RL Circuit:** A circuit consists of an inductance L and resistance R connected to a constant voltage V_0. At $t = 0$, the switch is closed. Solve $L\frac{dI}{dt} + RI = V_0$ with $I(0) = 0$ and identify the time constant.

2. **Newton's Law of Cooling:** A metal ingot at 800°C is placed in a quenching bath maintained at 25°C. After 2 minutes, the temperature is 400°C. Determine the time required for the temperature to drop to 50°C.

3. **Bernoulli Differential Equation:** Solve the nonlinear Bernoulli equation $\frac{dy}{dx} + \frac{1}{x}y = xy^2$ using the change of variable $u = y^{1-2} = y^{-1}$.

4. **Mixing Problem:** A tank initially contains 100 L of pure water. Saltwater with concentration 2 g/L flows in at rate 3 L/min, and well-mixed solution drains at 3 L/min. Write and solve the ODE for salt mass $m(t)$, and find the limiting salt concentration.

5. **Radioactive Chain Decay:** Nuclide A decays to B (decay constant λ_A), and B decays to stable C (λ_B). Given $N_A(0) = N_0$ and $N_B(0) = 0$, solve the coupled system to find $N_B(t)$. Show that N_B reaches a maximum at $t^* = \ln(\lambda_B/\lambda_A)/(\lambda_B - \lambda_A)$.



Chapter 9

Second-Order Linear Differential Equations & Oscillations

Chapter 9 Instructional Management Plan

1. Chapter Topics and Core Content

1. Homogeneous Linear Second-Order ODEs with Constant Coefficients
2. The Damped Harmonic Oscillator and Three Dynamic Regimes
3. Driven Oscillations, Resonance, and Quality Factor
4. Worked Examples: Shock Absorbers and RLC Resonance
5. Summary of Key Formulas and Chapter Exercises

2. Behavioral Learning Objectives (OBE)

1. Solve homogeneous second-order linear ODEs via characteristic auxiliary equations.
2. Analyze and physically classify the three damping regimes: overdamped, critically damped, and underdamped.
3. Calculate steady-state amplitude, phase lag, and resonance frequency in driven systems.
4. Quantify resonance sharpness and power dissipation using the Quality factor (Q).

3. Teaching Methods and Instructional Activities

1. Lecture on characteristic equations, complex roots, and phase portrait analysis.
2. Workshop deriving amplitude resonance curves and power absorption profiles.
3. Python computational laboratory simulating damped-driven oscillators and phase space spirals.
4. Historical case study on structural resonance (Tacoma Narrows Bridge collapse).

4. Instructional Media and Educational Technology

1. Course textbook: Chapter 9 Second-Order Linear Differential Equations & Oscillations.
2. Dynamic resonance curve animations and interactive phase space trajectories.
3. Python SciPy scripts for numerical integration of nonlinear driven pendulums.
4. Comprehensive problem sets and worked engineering application examples.

5. Measurement and Learning Assessment

1. Formative evaluation of in-class phase space trajectory classifications.
2. Homework assignment on driven oscillator amplitude response and Quality factor (Q).
3. Chapter 9 summative diagnostic examination on second-order ODEs and resonance.

9.1 Homogeneous Linear Second-Order ODEs with Constant Coefficients

Second-order differential equations are central to physics because Newton's second law $\vec{F} = m\frac{d^2\vec{x}}{dt^2}$, wave equations $\nabla^2\psi = \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2}$, and Schrödinger's equation are all of second order.

The standard homogeneous equation with real constant coefficients is:

$$a\frac{d^2x}{dt^2} + b\frac{dx}{dt} + cx = 0 \tag{9.1}$$

Substituting the trial solution $x(t) = e^{rt}$ yields the algebraic **characteristic equation**:

$$ar^2 + br + c = 0 \implies r_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{9.2}$$

9.2 The Damped Harmonic Oscillator and Three Dynamic Regimes

Consider a particle of mass m attached to a spring of constant k moving in a viscous damping medium with drag force $F_d = -b\dot{x}$. Newton's second law gives:

$$m\ddot{x} + b\dot{x} + kx = 0 \iff \ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0 \tag{9.3}$$

where $\omega_0 \equiv \sqrt{k/m}$ is the **undamped natural frequency**, and $\gamma \equiv \frac{b}{2m}$ is the **damping coefficient**. The characteristic roots are:

$$r_{1,2} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2} \tag{9.4}$$

Depending on the discriminant $\Delta = \gamma^2 - \omega_0^2$, the system exhibits three distinct physical regimes:

1. **Overdamped ($\gamma > \omega_0$):** Roots are real, distinct, and negative. The mass returns exponentially to equilibrium without oscillating:

$$x(t) = C_1 e^{-(\gamma-\sqrt{\gamma^2-\omega_0^2})t} + C_2 e^{-(\gamma+\sqrt{\gamma^2-\omega_0^2})t} \tag{9.5}$$

2. **Critically Damped ($\gamma = \omega_0$):** Repeated real root $r_1 = r_2 = -\gamma$. The system returns to equilibrium in the fastest possible time without overshoot:

$$x(t) = (C_1 + C_2 t)e^{-\gamma t} \tag{9.6}$$

This is the ideal design for automobile shock absorbers and analog galvanometer meters.

3. **Underdamped ($\gamma < \omega_0$):** Roots are complex conjugates $r_{1,2} = -\gamma \pm i\omega_d$, where $\omega_d \equiv \sqrt{\omega_0^2 - \gamma^2}$ is the **damped angular frequency**:

$$x(t) = Ae^{-\gamma t}\cos(\omega_d t - \phi) \tag{9.7}$$

The system oscillates with exponentially decaying amplitude envelopes $\pm Ae^{-\gamma t}$.

Regime	Condition	Characteristic Roots	Oscillatory?	Engineering Application
Overdamped	$\gamma > \omega_0$	Real, distinct: $r_1 \neq r_2 < 0$	No	Heavy industrial door closers
Critically Damped	$\gamma = \omega_0$	Real, repeated: $r = -\gamma$	No	Vehicle suspension, seismographs
Underdamped	$\gamma < \omega_0$	Complex: $-\gamma \pm i\omega_d$	Yes	Quartz clocks, atomic force microscopy

Table 9.1: Comparison of the three damping regimes in physical harmonic systems.

9.3 Driven Oscillations and Resonance

When subjected to an external sinusoidal driving force $F(t) = F_0 \cos(\omega t)$, the equation of motion becomes:

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = \frac{F_0}{m}\cos(\omega t) \tag{9.8}$$

The steady-state particular solution is:

$$x_p(t) = A(\omega)\cos(\omega t - \delta) \tag{9.9}$$

where the steady-state amplitude $A(\omega)$ and phase lag $\delta(\omega)$ are:

$$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}} \tag{9.10}$$

$$\delta(\omega) = \arctan\left(\frac{2\gamma\omega}{\omega_0^2 - \omega^2}\right) \tag{9.11}$$

Definition 9.1 Resonance Frequency and Quality Factor

The amplitude $A(\omega)$ reaches its absolute maximum at the **amplitude resonance frequency**:

$$\omega_r = \sqrt{\omega_0^2 - 2\gamma^2} \quad \text{(valid for } \gamma < \omega_0/\sqrt{2}\text{)} \tag{9.12}$$

The sharpness of the resonance curve is quantified by the dimensionless **Quality factor (Q)**:

$$Q \equiv \frac{\omega_0}{2\gamma} = \frac{\omega_0}{\Delta\omega_{\text{FWHM}}} \tag{9.13}$$

where $\Delta\omega_{\text{FWHM}}$ is the full width at half-maximum of the energy resonance profile.

Pitfall Alert: Distinction Between Three Characteristic Frequencies

Do not conflate the three distinct frequencies in damped systems:

- Undamped natural frequency: $\omega_0 = \sqrt{k/m}$.
- Damped natural frequency: $\omega_d = \sqrt{\omega_0^2 - \gamma^2} < \omega_0$.
- Amplitude resonance frequency: $\omega_r = \sqrt{\omega_0^2 - 2\gamma^2} < \omega_d < \omega_0$.

Only in the weakly damped limit ($\gamma \ll \omega_0$ or $Q \gg 1$) do these three frequencies converge: $\omega_0 \approx \omega_d \approx \omega_r$.

9.4 Worked Examples and Physical Applications

Example 9.1: Driven Oscillator Amplitude and Phase Response

An instrument package of mass $m = 2.0$ kg is mounted on springs with effective constant $k = 800$ N/m and damping $b = 4.0$ N · s/m. A harmonic shaker applies driving force $F_0 = 10$ N at frequency $\omega = 18$ rad/s. Calculate: (a) undamped frequency ω_0, (b) damping ratio γ, (c) quality factor Q, and (d) steady-state amplitude A and phase lag δ.

Picture / Conceptualization: The mass is driven harmonically near its natural resonance. Damping limits the peak amplitude.

Strategy / Governing Laws: Use the definitions $\omega_0 = \sqrt{k/m}$, $\gamma = b/(2m)$, and the steady-state response formulas for $A(\omega)$ and $\delta(\omega)$.

Calculation / Analytical Solution: 1. Natural frequency and damping parameter:

$$\omega_0 = \sqrt{\frac{800}{2.0}} = 20 \text{ rad/s} \tag{9.14}$$

$$\gamma = \frac{4.0}{2(2.0)} = 1.0 \text{ s}^{-1} \tag{9.15}$$

2. Quality factor:

$$Q = \frac{\omega_0}{2\gamma} = \frac{20}{2(1.0)} = 10 \tag{9.16}$$

3. Steady-state amplitude at $\omega = 18$ rad/s:

$$\omega_0^2 - \omega^2 = 20^2 - 18^2 = 400 - 324 = 76 \text{ (rad/s)}^2 \tag{9.17}$$

$$2\gamma\omega = 2(1.0)(18) = 36 \text{ (rad/s)}^2 \tag{9.18}$$

$$\text{Denominator} = \sqrt{(76)^2 + (36)^2} = \sqrt{5776 + 1296} = \sqrt{7072} \approx 84.1 \text{ (rad/s)}^2 \tag{9.19}$$

$$A(18) = \frac{F_0/m}{\text{Denominator}} = \frac{10/2.0}{84.1} = \frac{5.0}{84.1} \approx 0.0595 \text{ m} = 5.95 \text{ cm} \tag{9.20}$$

4. Phase lag:

$$\delta = \arctan\left(\frac{36}{76}\right) = \arctan(0.4737) \approx 0.442 \text{ rad} = 25.3^\circ \tag{9.21}$$

Think / Physical Insights: Because $\omega < \omega_0$, the driving frequency is below resonance, and the response lags by less than $90°$ ($\delta = 25.3°$). At exact resonance ($\omega = \omega_0 = 20$ rad/s), the phase lag would be exactly $90°$, and amplitude would reach $A_{\text{max}} \approx \frac{F_0/m}{2\gamma\omega_0} = \frac{5.0}{40} = 0.125$ m $= 12.5$ cm. ■

Driven Harmonic Resonance Curves in Python

```
import numpy as np
import matplotlib.pyplot as plt

omega_0 = 20.0
F0_m = 5.0
omega = np.linspace(5, 35, 1000)

for Q in [2, 5, 10, 20]:
    gamma = omega_0 / (2 * Q)
    A = F0_m / np.sqrt((omega_0**2 - omega**2)**2 + 4 * gamma**2 *
    omega**2)
    print(f"Q = {Q:2d}: Peak Amplitude = {np.max(A)*100:.2f} cm")
```

Example 9.2: Driven RLC Resonance and Q-Factor Analysis

A series RLC circuit has $L = 10$ mH, $C = 100$ μF, $R = 2$ Ω, and is driven by $\mathcal{E}(t) = 12\cos(\omega t)$ V. (i) Find the resonance frequency ω_0, (ii) compute the Q-factor and bandwidth $\Delta\omega$, (iii) calculate the maximum current amplitude at resonance, and (iv) identify the analogous mechanical oscillator.

Solution:

(i) Resonance frequency:

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10 \times 10^{-3} \times 100 \times 10^{-6}}} = \frac{1}{\sqrt{10^{-6}}} = 1000 \text{ rad/s} \tag{9.22}$$

Corresponding to $f_0 = \omega_0/2\pi \approx 159.2$ Hz.

(ii) Q-factor and bandwidth:

$$Q = \frac{\omega_0 L}{R} = \frac{1000 \times 0.010}{2} = \boxed{5} \qquad \Delta\omega = \frac{\omega_0}{Q} = \frac{1000}{5} = 200 \text{ rad/s} \tag{9.23}$$

The $Q = 5$ indicates moderate selectivity: at frequencies $|\omega - \omega_0| > 100$ rad/s, the response amplitude falls below $1/\sqrt{2}$ of peak.

(iii) Maximum current at resonance (where $X_L = X_C$ cancel):

$$\tilde{Z}(\omega_0) = R + i(\omega_0 L - 1/\omega_0 C) = R + 0 = 2\ \Omega \implies \boxed{I_{\text{max}} = \frac{\mathcal{E}_0}{R} = \frac{12}{2} = 6 \text{ A}} \tag{9.24}$$

(iv) Mechanical analogy: The RLC equation $L\ddot{Q} + R\dot{Q} + Q/C = \mathcal{E}_0 \cos\omega t$ maps exactly to $m\ddot{x} + b\dot{x} + kx = F_0 \cos\omega t$ via: $L \leftrightarrow m$, $R \leftrightarrow b$, $1/C \leftrightarrow k$, $Q \leftrightarrow x$, $\mathcal{E}_0 \leftrightarrow F_0$.

Think / Physical Insights: The Q-factor simultaneously quantifies: (1) energy stored relative to energy dissipated per cycle, $Q = 2\pi E/\Delta E_{\text{cycle}}$; (2) frequency selectivity (bandwidth); and (3) oscillation longevity after the drive is switched off ($\tau_{\text{decay}} = Q/\omega_0$). High-$Q$ resonators (quartz crystal, optical cavities) are fundamental to precise timekeeping and spectroscopy. ■

9.5 Summary of Key Formulas

Oscillator Parameter	Formula	Physical Meaning
Natural Frequency	$\omega_0 = \sqrt{k/m}$	Oscillation frequency without damping or driving
Damped Frequency	$\omega_d = \sqrt{\omega_0^2 - \gamma^2}$	Actual oscillation rate under viscous resistance
Amplitude Resonance	$\omega_r = \sqrt{\omega_0^2 - 2\gamma^2}$	Frequency of maximum steady-state displacement response
Quality Factor (Q)	$Q = \frac{\omega_0}{2\gamma} = 2\pi\frac{E}{\Delta E_{\text{cycle}}}$	Measure of energy retention per cycle and frequency selectivity
RLC Resonance	$\omega_0 = 1/\sqrt{LC}$, $Q = \omega_0 L/R$	Electrical analogue of the mechanical oscillator

Table 9.2: Key relationships for damped and driven harmonic oscillators.

9.6 Chapter Exercises

1. **Underdamped Logarithmic Decrement:** Show that the ratio of two successive maximum displacements x_n and x_{n+1} of an underdamped oscillator separated by period $T = 2\pi/\omega_d$ satisfies $\delta_{\text{log}} \equiv \ln(x_n/x_{n+1}) = \gamma T = \frac{2\pi\gamma}{\sqrt{\omega_0^2-\gamma^2}}$.

2. **Bridge Resonance Safety:** An army detachment marches across a suspension bridge of mass $M = 50,000$ kg and natural frequency $f_0 = 1.2$ Hz. Explain why troops are ordered to break step using the resonance amplitude formula.

3. **Electrical-Mechanical Analogies:** Construct a dictionary table mapping mass m, damping b, spring k, position x, and driving force F_0 to their electrical series LRC circuit counterparts.

4. **Coupled Pendulums:** Two identical pendulums of length ℓ are connected by a spring of constant k at their midpoints. Let θ_1 and θ_2 be the angular displacements. Write the linearised equations of motion, introduce normal coordinates $q_\pm = \theta_1 \pm \theta_2$, and identify the two normal mode frequencies $\omega_\pm$.

5. **Critically Damped Shock Absorber:** A car's suspension system has spring constant $k = 12$ kN/m and effective mass $m = 300$ kg. Calculate the critical damping coefficient $b_c = 2\sqrt{mk}$ needed to eliminate oscillation after a road bump, and write the general solution $x(t)$ for critical damping with initial conditions $x(0) = x_0$, $\dot{x}(0) = 0$.

Chapter A

Mathematical Formulas & Vector Identities

A.1 Differential Operators in Curvilinear Coordinates

Let ψ denote a smooth scalar field and $\vec{A} = A_1\hat{e}_1 + A_2\hat{e}_2 + A_3\hat{e}_3$ denote a continuously differentiable vector field.

A.1.1 Cartesian Coordinates (x, y, z)

Metric scale factors: $h_x = 1,\ h_y = 1,\ h_z = 1.$

$$\nabla\psi = \frac{\partial\psi}{\partial x}\hat{i} + \frac{\partial\psi}{\partial y}\hat{j} + \frac{\partial\psi}{\partial z}\hat{k} \tag{A.1}$$

$$\nabla\cdot\vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \tag{A.2}$$

$$\nabla\times\vec{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{i} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{j} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{k} \tag{A.3}$$

$$\nabla^2\psi = \frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} \tag{A.4}$$

A.1.2 Cylindrical Coordinates (ρ, ϕ, z)

Metric scale factors: $h_\rho = 1,\ h_\phi = \rho,\ h_z = 1.$

$$\nabla\psi = \frac{\partial\psi}{\partial\rho}\hat{\rho} + \frac{1}{\rho}\frac{\partial\psi}{\partial\phi}\hat{\phi} + \frac{\partial\psi}{\partial z}\hat{k} \tag{A.5}$$

$$\nabla\cdot\vec{A} = \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho A_\rho) + \frac{1}{\rho}\frac{\partial A_\phi}{\partial\phi} + \frac{\partial A_z}{\partial z} \tag{A.6}$$

$$\nabla\times\vec{A} = \left(\frac{1}{\rho}\frac{\partial A_z}{\partial\phi} - \frac{\partial A_\phi}{\partial z}\right)\hat{\rho} + \left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial\rho}\right)\hat{\phi} + \frac{1}{\rho}\left(\frac{\partial(\rho A_\phi)}{\partial\rho} - \frac{\partial A_\rho}{\partial\phi}\right)\hat{k} \tag{A.7}$$

$$\nabla^2\psi = \frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial\psi}{\partial\rho}\right) + \frac{1}{\rho^2}\frac{\partial^2\psi}{\partial\phi^2} + \frac{\partial^2\psi}{\partial z^2} \tag{A.8}$$

A.1.3 Spherical Coordinates (r, θ, ϕ)

Metric scale factors: $h_r = 1,\ h_\theta = r,\ h_\phi = r\sin\theta$.

$$\nabla\psi = \frac{\partial\psi}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial\psi}{\partial\theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial\psi}{\partial\phi}\hat{\phi} \tag{A.9}$$

$$\nabla\cdot\vec{A} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 A_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta A_\theta) + \frac{1}{r\sin\theta}\frac{\partial A_\phi}{\partial\phi} \tag{A.10}$$

$$\begin{aligned}\nabla\times\vec{A} &= \frac{1}{r\sin\theta}\left(\frac{\partial(\sin\theta A_\phi)}{\partial\theta} - \frac{\partial A_\theta}{\partial\phi}\right)\hat{r} \\ &\quad + \frac{1}{r}\left(\frac{1}{\sin\theta}\frac{\partial A_r}{\partial\phi} - \frac{\partial(rA_\phi)}{\partial r}\right)\hat{\theta} + \frac{1}{r}\left(\frac{\partial(rA_\theta)}{\partial r} - \frac{\partial A_r}{\partial\theta}\right)\hat{\phi}\end{aligned} \tag{A.11}$$

$$\nabla^2\psi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2} \tag{A.12}$$

A.2 Fundamental Vector and Differential Identities

$$\vec{A}\cdot(\vec{B}\times\vec{C}) = \vec{B}\cdot(\vec{C}\times\vec{A}) = \vec{C}\cdot(\vec{A}\times\vec{B}) \tag{A.13}$$

$$\vec{A}\times(\vec{B}\times\vec{C}) = (\vec{A}\cdot\vec{C})\vec{B} - (\vec{A}\cdot\vec{B})\vec{C} \tag{A.14}$$

$$\nabla\cdot(\nabla\times\vec{A}) = 0 \quad \text{(Divergence of a curl is zero)} \tag{A.15}$$

$$\nabla\times(\nabla\psi) = \vec{0} \quad \text{(Curl of a gradient is zero)} \tag{A.16}$$

$$\nabla\times(\nabla\times\vec{A}) = \nabla(\nabla\cdot\vec{A}) - \nabla^2\vec{A} \tag{A.17}$$

$$\nabla(\vec{A}\cdot\vec{B}) = \vec{A}\times(\nabla\times\vec{B}) + \vec{B}\times(\nabla\times\vec{A}) + (\vec{A}\cdot\nabla)\vec{B} + (\vec{B}\cdot\nabla)\vec{A} \tag{A.18}$$

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Subject Index

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Educational Background

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- Computational Physics & Scientific Python

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- **Computational Physics:** High-performance scientific numerical simulations and Python modeling.
- **Immersive Educational Technology:** Augmented, Virtual, and Extended Reality (AR/VR/XR) in STEM.
- **Artificial Intelligence in Education:** Machine learning and AI-assisted physics learning environments.